

**CEBU INSTITUTE OF TECHNOLOGY  
UNIVERSITY**

**COLLEGE OF COMPUTER STUDIES**

**Software Requirements Specifications**

*for*  
IntelliTrack

## Change History

# Table of Contents

|                                              |          |
|----------------------------------------------|----------|
| <b>Change History</b>                        | <b>2</b> |
| <b>Table of Contents</b>                     | <b>3</b> |
| <b>1. Introduction</b>                       | <b>4</b> |
| 1.1. Purpose                                 | 4        |
| 1.2. Scope                                   | 4        |
| 1.3. Definitions, Acronyms and Abbreviations | 4        |
| 1.4. References                              | 4        |
| <b>2. Overall Description</b>                | <b>5</b> |
| 2.1. Product perspective                     | 5        |
| 2.2. User characteristics                    | 5        |
| 2.4. Constraints                             | 5        |
| 2.5. Assumptions and dependencies            | 6        |
| <b>3. Specific Requirements</b>              | <b>7</b> |
| 3.1. External interface requirements         | 7        |
| 3.1.1. <i>Hardware interfaces</i>            | 7        |
| 3.1.2. <i>Software interfaces</i>            | 7        |
| 3.1.3. <i>Communications interfaces</i>      | 7        |
| 3.2. Functional requirements                 | 7        |
| <i>Module 1</i>                              | 7        |
| <i>Module 2</i>                              | 8        |
| 3.4. Non-functional requirements             | 8        |
| <i>Performance</i>                           | 8        |
| <i>Security</i>                              | 8        |
| <i>Reliability</i>                           | 8        |

# 1. Introduction

## 1.1. Purpose

Capstone projects often fail due to students' poor time management, disorganized submissions, and last-minute work across multiple deliverables. Faculty advisers also struggle with manually tracking various groups, monitoring deadlines, reviewing submissions, and providing timely feedback. Existing platforms such as LMSs and academic portals offer only basic file submission and storage. They lack milestone-based tracking, automated reminders, version control, risk detection, and structured feedback loops specifically designed for capstone workflows.

The purpose of the **IntelliTrack** is to provide an AI-powered submission tracker that automates deadline monitoring, predicts late submissions, manages version histories, and delivers real-time dashboards for students, advisers, coordinators, and administrators. This system enhances accountability, streamlines the entire submission workflow, improves adviser feedback processes, and ensures data-driven monitoring of capstone deliverables.

## 1.2. Scope

The **IntelliTrack** is a specialized, web-based, AI-powered platform designed to streamline the submission, monitoring, and evaluation of capstone project deliverables. Included features and functionalities are:

- Role-Based User Management (StudentID)
- Admin Panel Automated Submission Tracking & AI-Powered Reminders
- Document Management & Feedback System
- Real-Time Analytics & Reporting Dashboard
- Administration & Configuration
- Security & Privacy Framework

### 1.3. Definitions, Acronyms and Abbreviations

| Acronym & Abbreviations | Definition                                                                                               |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| AI                      | Artificial Intelligence                                                                                  |
| RBAC                    | Role-Based Access Control                                                                                |
| LMS                     | Learning Management System                                                                               |
| UI                      | User Interface                                                                                           |
| UX                      | User Experience                                                                                          |
| CRUD                    | Create, Read, Update, Delete (basic operations in software systems)                                      |
| CDSS                    | Capstone Deliverables Submission System                                                                  |
| Milestone               | A required capstone deliverable with a specific deadline (e.g., Proposal, Midterm, Final)                |
| Version Control         | A system that tracks document versions, changes, and submission histories                                |
| Analytics Dashboard     | A visual interface showing real-time submission status and progress insights                             |
| Inline Feedback         | Comments or annotations placed directly on a student's submitted document                                |
| Adviser                 | Faculty member assigned to a student group to evaluate submissions and provide structured feedback       |
| Coordinator             | Faculty member who manages capstone deadlines, oversees multiple groups, and generates academic reports  |
| Student                 | Capstone group member responsible for uploading deliverables and complying with deadlines                |
| Administrator (Admin)   | User responsible for system maintenance, user account management, system settings, and platform security |

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## 2. Overall Description

### 2.1. Product perspective

The **IntelliTrack** is a web-based platform designed to streamline the management, tracking, and evaluation of capstone project submissions. It follows a client–server architecture, where users access the system through a browser while backend services handle authentication, document storage, version control, analytics, and AI-driven reminders.

The product integrates with external services such as:

- **Institutional email system** for account verification
- **Cloud storage services** for secure file handling
- **AI modules** for risk prediction and smart reminders

**IntelliTrack** operates as a standalone system but can be embedded into an institution's IT ecosystem. It supports multiple roles (Students, Advisers, Coordinators, Administrators) and centralizes all capstone-related workflows in one structured platform.

### 2.2. User characteristics

#### Students

- Upload deliverables per milestone
- Receive automated and AI-generated deadline reminders
- Track submission status and version history
- View adviser feedback and revise documents accordingly

#### Advisers

- Review submissions
- Provide structured feedback, inline comments, and rubric-based evaluation

- Monitor student progress through dashboards
- Access version histories for transparency

## **Coordinators**

- Set deadlines, manage milestones, and oversee multiple project groups
- Approve submissions when required
- Generate class-wide or department-level reports
- Monitor adviser responsiveness and group compliance
- Monitor student progress through dashboards
- Access version histories for transparency

## **Administrators**

- Manage user accounts, roles, and permissions
- Configure system settings, academic periods, and platform rules
- Oversee system performance, security, and data integrity
- Maintain audit logs and ensure compliance with institutional policies

## **2.4. Constraints**

The development and deployment of **IntelliTrack** are subject to the following constraints:

### **Regulatory Policies**

- Must comply with institutional academic policies
- Must follow data privacy regulations such as Data Privacy Act of 2012 (Philippines)
- FERPA/GDPR alignment when applicable

### **Hardware & Software Limitations**

- Requires stable internet access
- Dependent on availability of cloud hosting



- Performance may vary with large file uploads

## **Interface Constraints**

- Must integrate with institutional email systems
- Must support document formats (PDF, DOCX, PPTX)
- Connection with cloud storage API

## **Parallel Operation**

- Must support simultaneous access from multiple users (students, advisers, admins)
- Should maintain performance at 500-1000 and more concurrent users

## **Audit & Logging Requirements**

- Must maintain audit trails for all submissions, edits, role changes, and feedback
- Logs should be immutable for academic integrity

## **Control Functions**

- Version control must prevent document overwriting
- System must enforce role-based restrictions

## **Reliability Requirements**

- Target uptime: **99.9% availability**
- Automated backups and recovery plans required

## **Criticality**

- High academic impact: failure may affect grading and submission deadlines
- Must ensure reliable access, especially near milestone deadlines

## **Safety & Security Considerations**

- Encrypted document storage
- Role-based access control
- Anomaly detection for suspicious activity
- Secure authentication for faculty accounts

## **2.5. Assumptions and dependencies**

### **Assumptions**

The following conditions are assumed to be true for the **IntelliTrack** to operate as intended:

- Users have stable internet access and use modern browsers (Chrome, Edge, Firefox, Safari).
- Institutional administrators will provide accurate academic schedules, deadlines, user lists, and role assignments.
- Students and faculty are familiar with uploading and downloading digital documents.
- Institutional email services are available and functioning for authentication and account verification.
- Cloud hosting services (e.g., Azure, AWS) remain operational and available.
- All users access the system with their assigned institution-based accounts.
- Advisers and coordinators provide timely feedback and updates to ensure analytics remain accurate.
- Document uploads follow accepted formats (PDF, DOCX, PPTX) as defined by institutional policies.
- Users have devices capable of accessing the system (PC, laptop, or mobile device).
- File sizes uploaded by students are within allowed storage limitations.
- System maintenance schedules are coordinated and communicated in advance to avoid academic disruptions.
- Institutional policies regarding data retention, privacy, and academic conduct remain stable throughout development.

## Dependencies

The **IntelliTrack** relies on the following external systems, technologies, and conditions:

- Availability and reliability of the cloud database (e.g., Supabase) for storing submissions, logs, and metadata.
- AI-generated reminders and risk detection depend on accurate and complete data inputs (deadlines, submission timestamps, user activity).
- Document management, version control, and rollback features rely on third-party storage APIs and cloud file systems.
- Real-time analytics and reporting depend on uninterrupted logging services and complete submission records.
- Security features depend on the hosting provider's infrastructure for encryption, secure authentication, and intrusion detection.
- Email notification services must remain available.
- Integration with institutional systems (e.g., student information systems, email servers) must remain functional.
- System performance depends on the scalability and responsiveness of the hosting platform.
- Mobile access depends on device compatibility and browser support for responsive UI components.
- Backup and recovery procedures depend on cloud provider backup services.
- Future upgrades may depend on API updates or changes from third-party providers.
- The development team relies on version control tools (e.g., GitHub) for stable collaboration and deployment pipelines.

## 3. Specific Requirements

### 3.1. External interface requirements

#### 3.1.1. Hardware interfaces

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##### Client Devices:

- **Devices Supported:** The system must be accessible on any device with a modern browser, including PC, laptop, or mobile device (though the UI/UX is tailored for desktop compatibility).
- **User Interface (UI) Support:** The system requires full-screen support and must render a simple, intuitive interface with easy navigation across different screen resolutions (responsive design is implied for cross-platform compatibility).
- **Input/Output:** Standard input devices (keyboard, mouse/touchpad) are required for login, document upload, and inputting inline comments/rubric evaluations.

##### Deployment & Hosting:

- **Server Hardware:** The system requires Cloud hosting with database support.
- **Performance Configuration:** The underlying server hardware must be provisioned to handle at least 1,000 concurrent users and maintain a response time of < 2 seconds.

#### 3.1.2. Software interfaces

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##### Required Software Products (Core Stack):

- **Operating System:** Any standard operating system that supports the required modern web browsers.
- **Data Management System:** The system is dependent on the availability and reliability of a cloud database for structured storage of submissions, logs, and metadata.
- **Backend Framework:** Django is required for secure, modular business logic.
- **Frontend Framework:** React.jsx is required for the user interface.
- **Version Control:** The development team relies on **version control tools (e.g.,**

**GitHub)** for stable collaboration.

### Possible Interfaces with Other Application Systems:

- **Institutional Email System:** The system must **integrate with institutional email systems** for **account verification**, login, and sending **automated reminders (via email)**.
- **Cloud Storage Services:** The system must connect with **cloud storage API** for **secure file handling**, managing **document version control**, and **rollback features**.
- **AI/Analytics Modules:** Interfaces with **AI modules** for **risk prediction**, **smart reminders**, and **analytics engines** for **progress tracking and reporting**.

#### 3.1.3. *Communications interfaces*

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**Network Access:** The system requires **stable internet access** as it is a web-based, cloud-hosted application.

#### **Protocols and Security:**

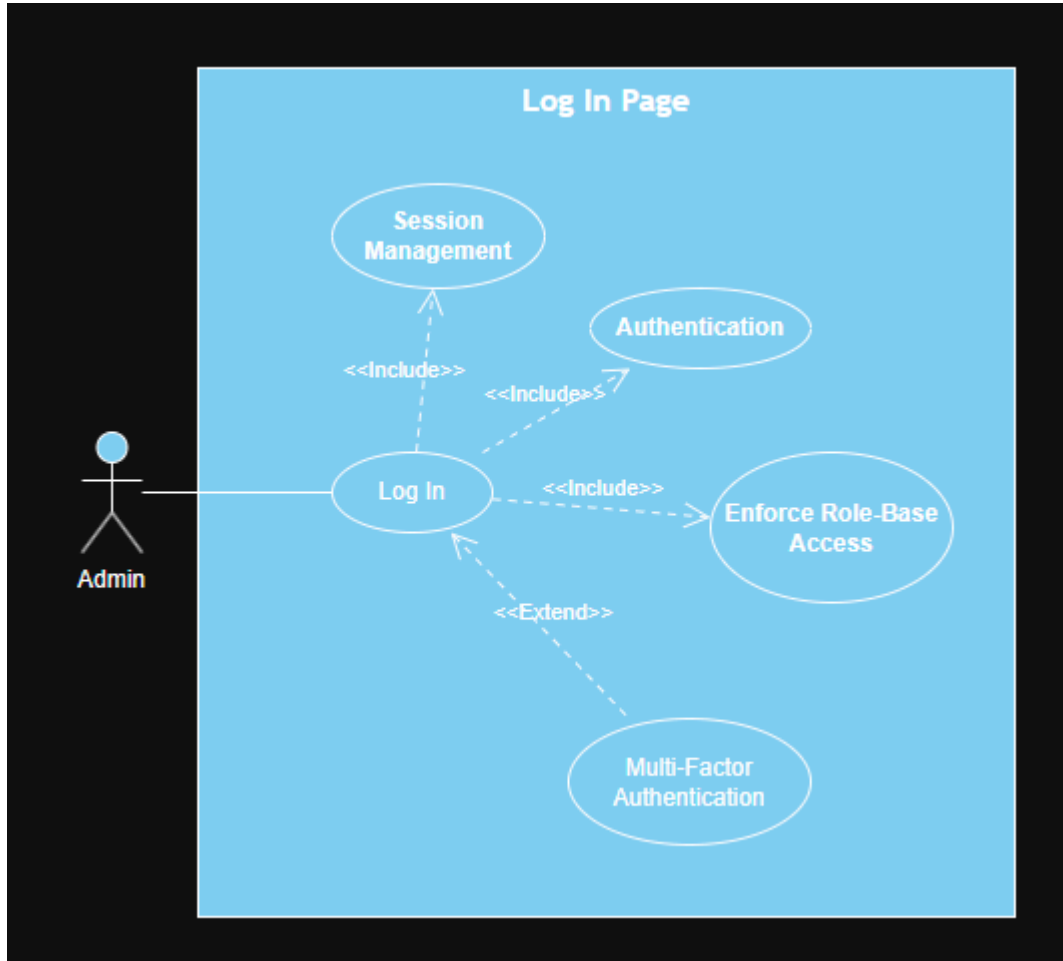
- **Data Encryption:** Requires protocols to ensure **end-to-end encryption for stored and transmitted data**.
- **Authentication Protocol:** Must integrate with institutional authentication protocols to use **institutional email** and support **Multi-Factor Authentication (MFA)** for advisers and administrators.
- **Email Protocol:** The system relies on **Email notification services (SMTP or third-party services)** for sending alerts and reminders.
- **API Communication:** Communication protocols are needed for the **connection with cloud storage API** and other external services (analytics, institutional systems).

### 3.2. *Functional requirements*

## Module 1: User Management System

### 1.1 Login Page

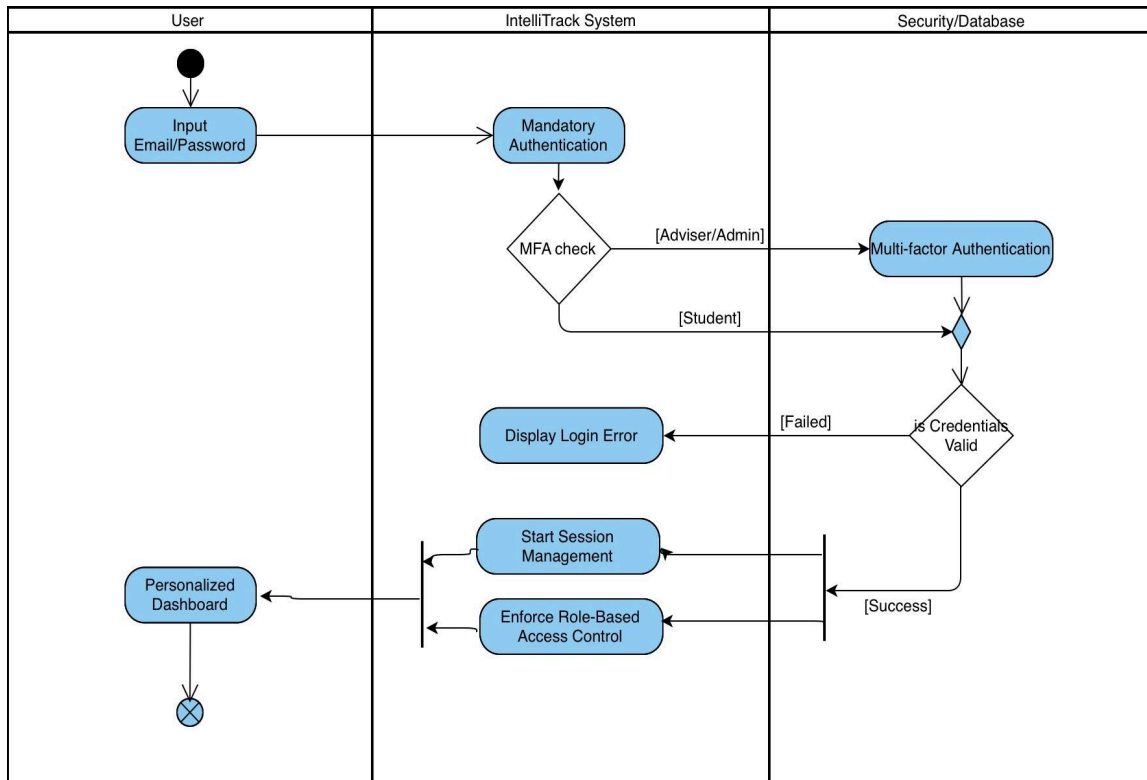
- Use Case Diagram



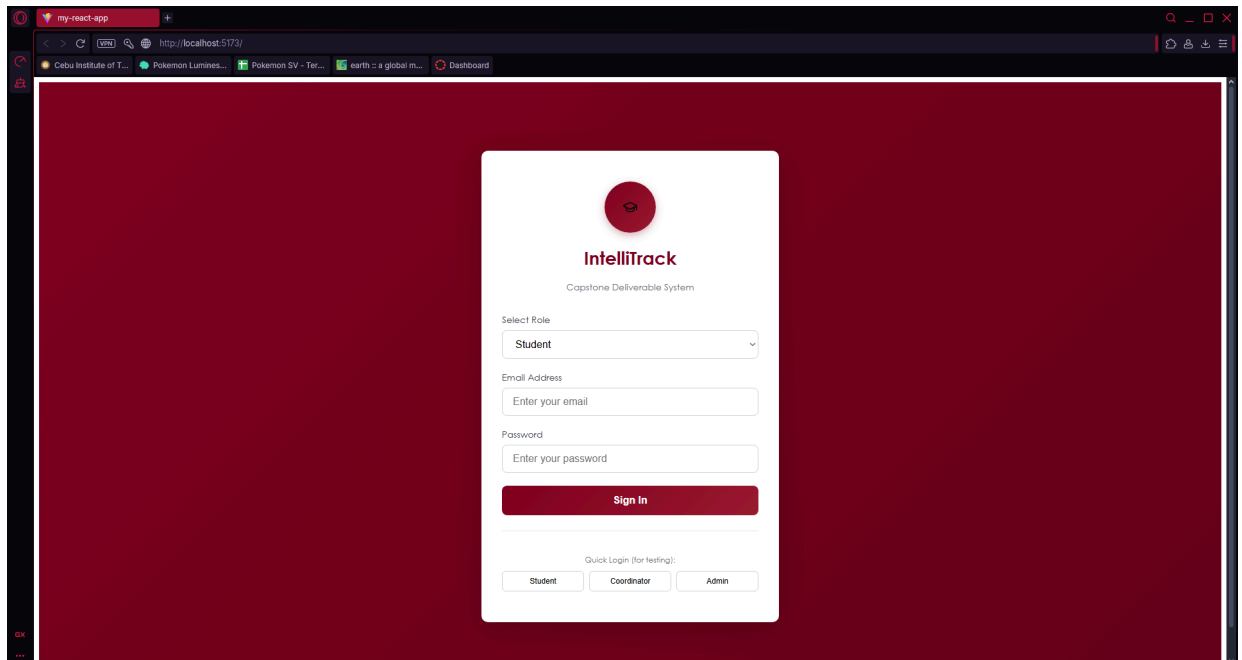
- Use Case Description

The Login use case serves as the foundational security step for IntelliTrack, initiated when the User (who can be a Student, Adviser, or Administrator) accesses the application and inputs their credentials. This action immediately triggers the mandatory Authentication process, where the system verifies the provided institutional email and password against stored records. For Advisers and Administrators, the system may require an additional security layer via Multi-Factor Authentication (MFA). Upon successful verification, the system begins Session Management and enforces Role-Based Access Control (RBAC) before directing the user to their personalized Home Page/Dashboard.

- Activity Diagram

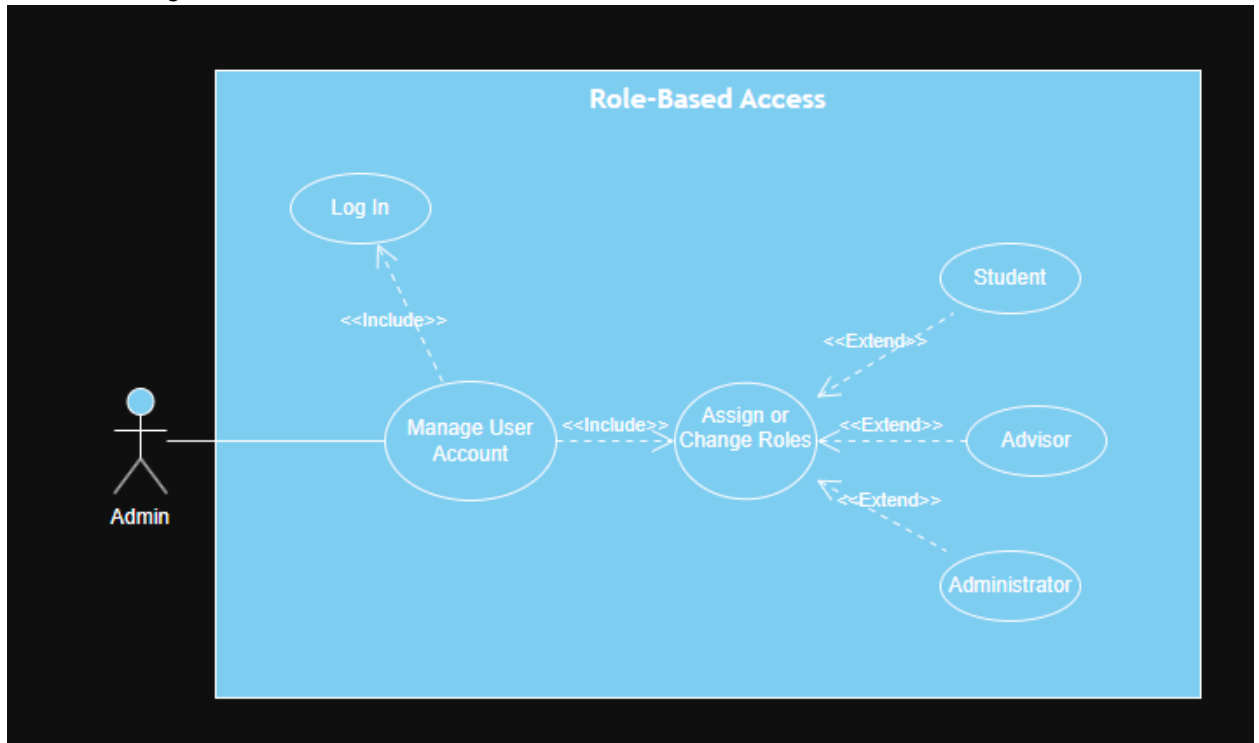


- Wireframe



## 1.2 Role-Based Access

- Use Case Diagram

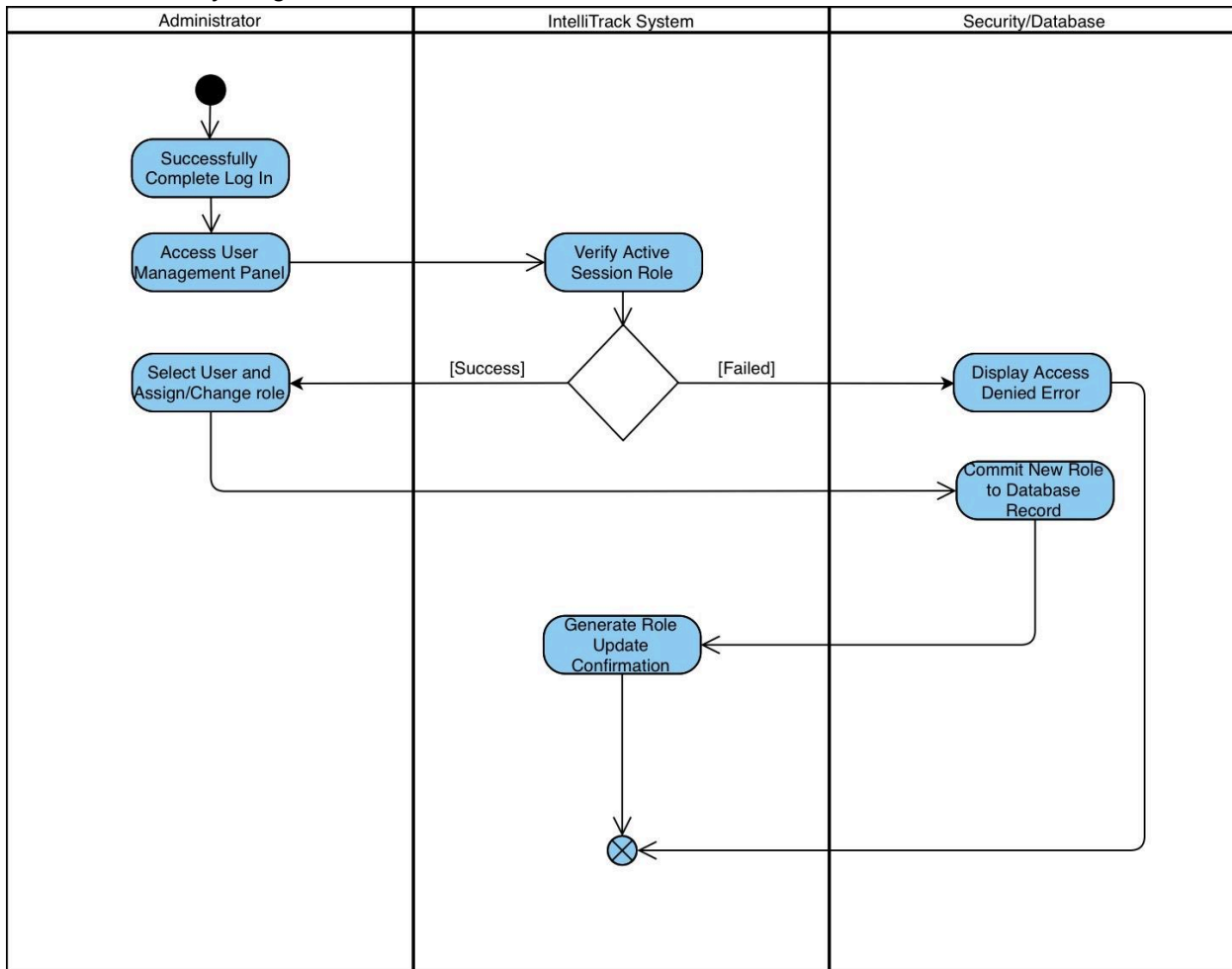


- Use Case Description

The Manage User Account use case represents the administrative core capability exclusively reserved for the Admin role, dictating the system's role-based access structure. Initiating this function first requires the Admin to successfully complete the Log In process to ensure security and auditability. Crucially, this use case includes the Assign or Change Roles function, allowing the Admin to designate a user as a Student, Adviser, or another Admin as needed. This central control mechanism is essential for system governance, ensuring all user permissions are correctly assigned and maintained by the highest authority. This explicit structure guarantees that no other user role can interfere with system access controls.



- Activity Diagram



- Wireframe

Admin

Capstone Management System - Admin

3

Logout

Admin User  
Admin

Home

User Management

System Config

Deadlines

Analytics

Welcome, Admin User!

Administrator Dashboard - Manage the entire capstone system

Total Students  
156

Total Advisers  
8

Active Projects  
12

Pending Reviews  
24

Recent System Activity

Latest actions and events in the system

New submission from Group 5  
5 minutes ago

Adviser Dr. Smith approved Group 2 proposal  
1 hour ago

New user registered: John Doe  
2 hours ago

Deadline reminder sent to all students  
3 hours ago

Coordinator

Capstone Management System - Coordinator

3

Logout

Dr. Jane Smith  
Coordinator

Home

Calendar

Profile

Feedback Evaluation

Welcome back, Dr. Jane Smith!

Overview of student projects and submissions

Advised Teams  
3

Pending Reviews  
8

Reviewed This Week  
12

At-Risk Projects  
1

Pending Reviews

Student submissions awaiting your feedback

| STUDENT      | DOCUMENT         | SUBMITTED ON | PRIORITY | ACTION |
|--------------|------------------|--------------|----------|--------|
| John Doe     | Project Proposal | 2025-12-01   | high     | Review |
| Jane Smith   | SRS Document     | 2025-12-02   | medium   | Review |
| Mike Johnson | SDD Document     | 2025-12-03   | low      | Review |

Advised Teams

Teams under your supervision

Student

Capstone Management System - Student

3

Logout

John Doe  
Student

Home

Submissions

Profile

Welcome back, John Doe!

Here's an overview of your capstone project progress

Total Submissions  
8

Pending Reviews  
2

Approved  
5

Revisions Needed  
1

Upcoming Deadlines

Stay on track with your submissions

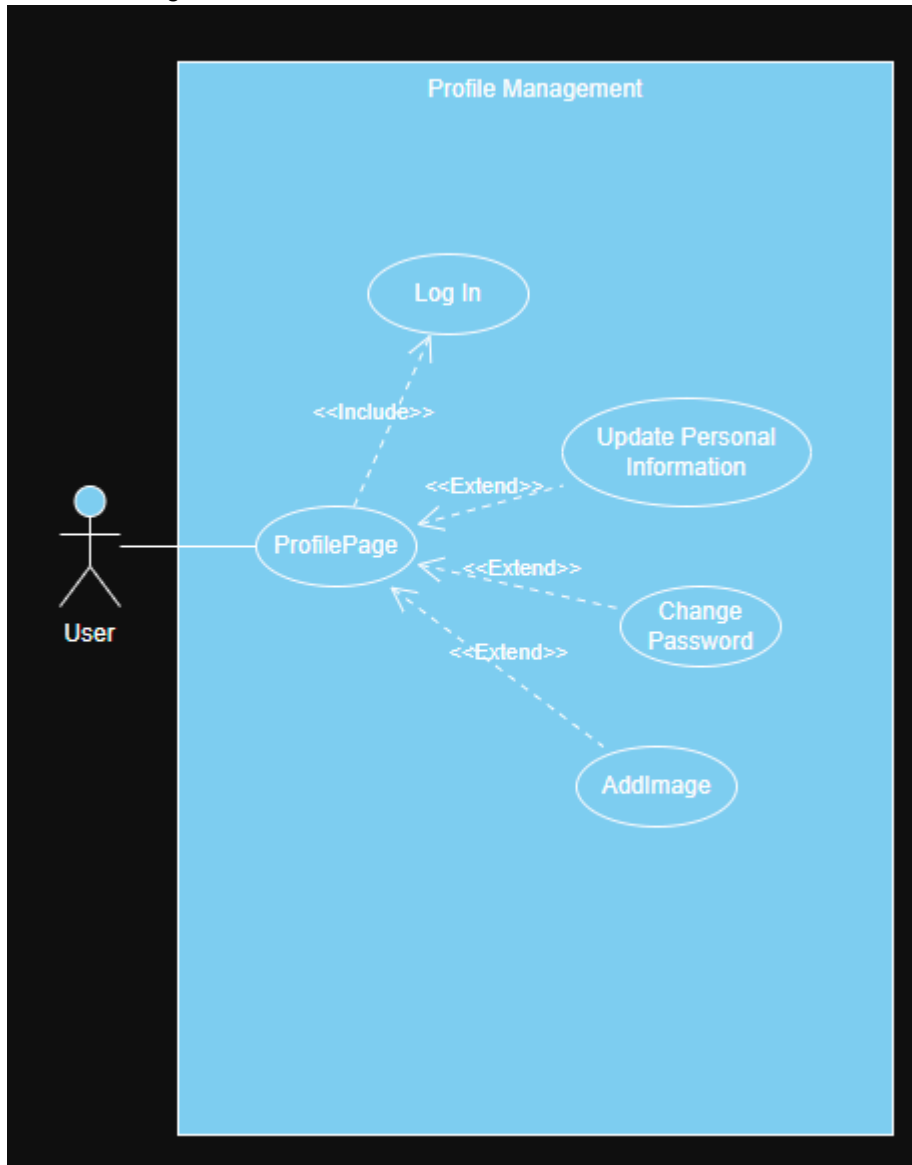
| DOCUMENT         | DUE DATE   | STATUS      | ACTION       |
|------------------|------------|-------------|--------------|
| Project Proposal | 2025-12-15 | pending     | View Details |
| SRS Document     | 2025-12-20 | in progress | View Details |
| SDD Document     | 2025-12-28 | not started | View Details |

Recent Submissions

Your latest submitted work

### 1.3 Profile Management

- Use Case Diagram

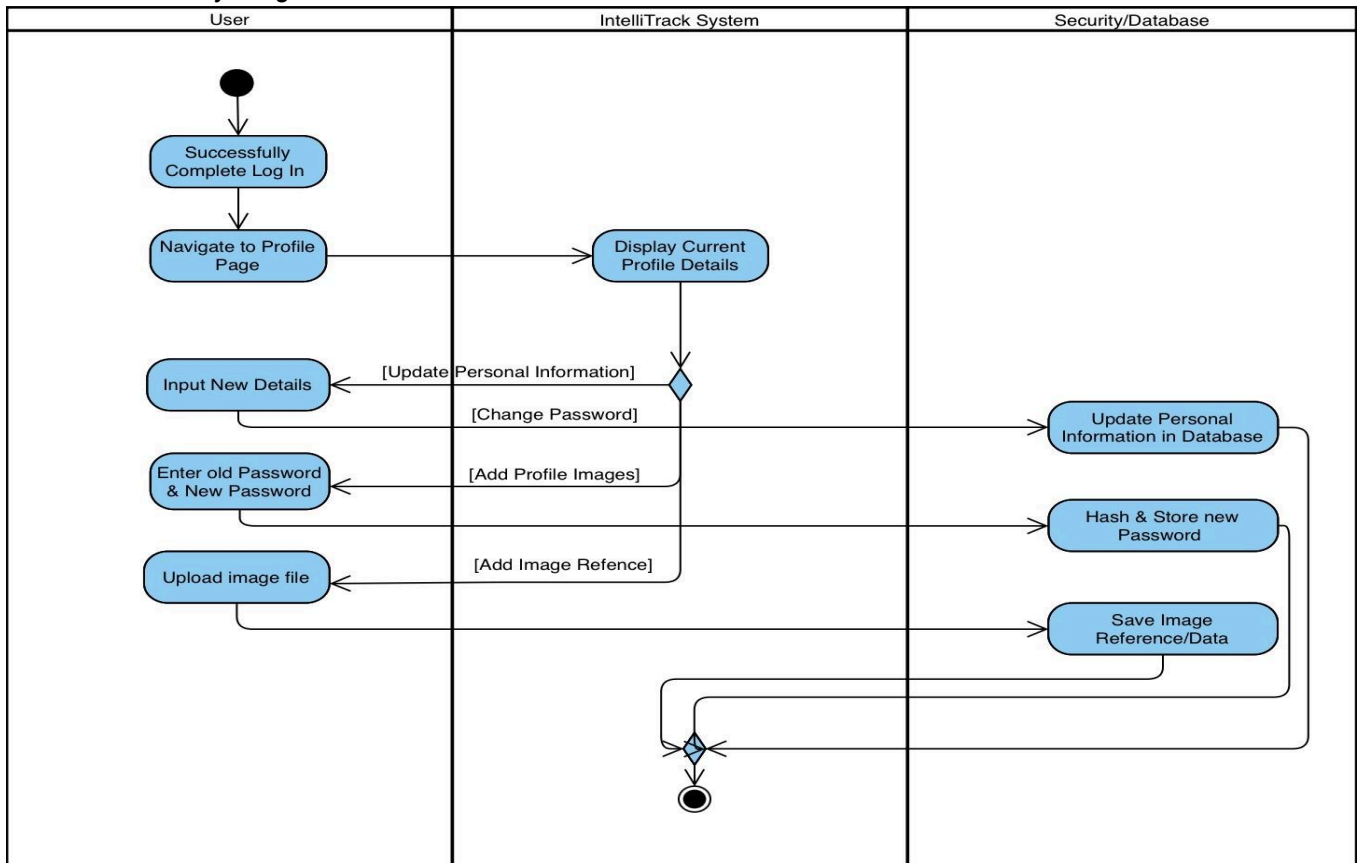


- Use Case Description

The User interacts directly with the central ProfilePage function to manage their account. To ensure system security and personalization, accessing the ProfilePage requires the successful completion of the Log In process. Once authenticated, the user views their personal profile details and is provided with several options for modifying their information. These options include dedicated functions to Update Personal Information (such as changing name or email), Change Password (for updating security credentials), and Add Image (for uploading or modifying a profile

picture). All of these profile modification actions are performed as specialized tasks triggered from the main profile view.

- Activity Diagram



• Wireframe

Capstone Management System - Student

John Doe

Student

Logout

Home

Submissions

Profile

My Profile

View and manage your profile information

Personal Information

John Doe

Student

Edit Profile

Full Name

John Doe

Email Address

student@example.com

Student ID

2024-00001

Program

Bachelor of Science in Computer Science

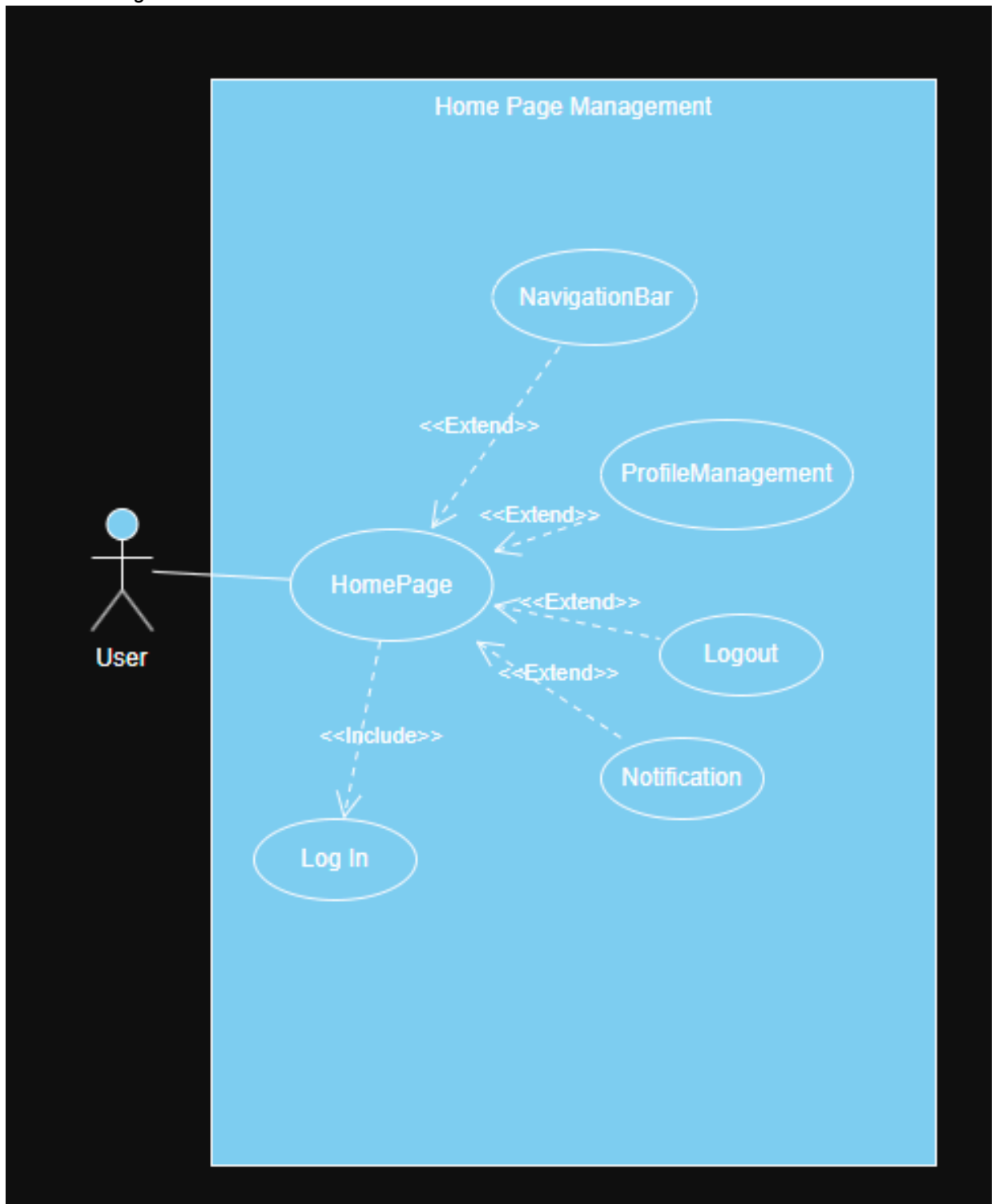
Year Level

4th Year

Capstone Project Details

## 1.4 Home Page

- Use Case Diagram

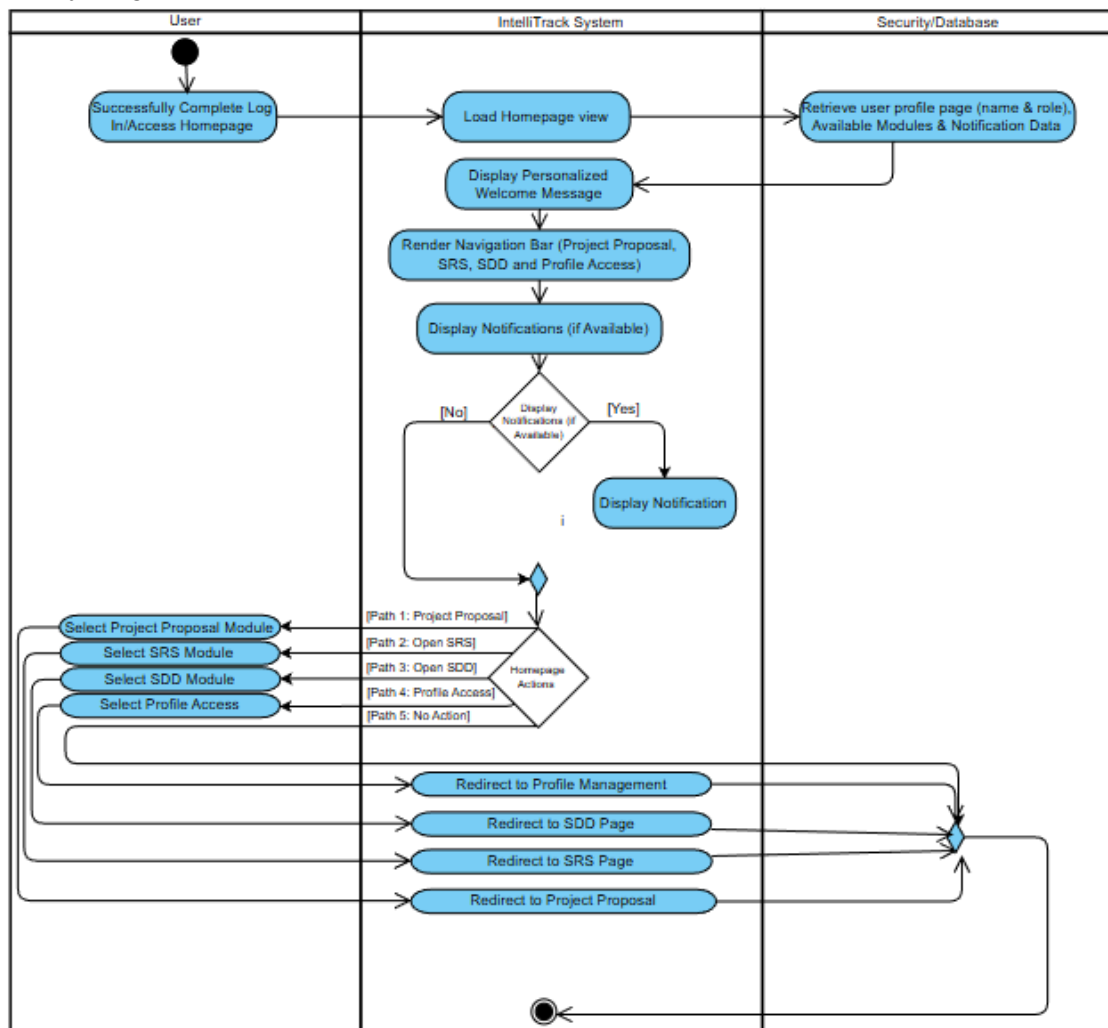


- Use Case Description

The HomePage serves as the central operational hub for the IntelliTrack

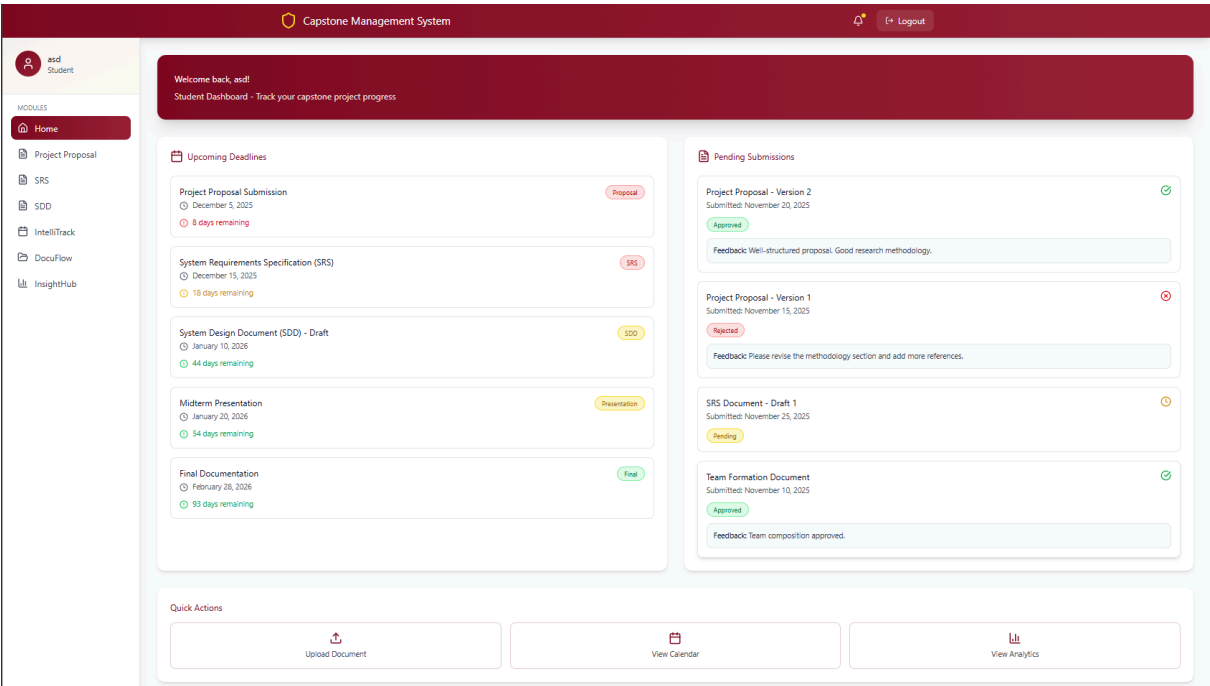
system, providing the User with a personalized dashboard immediately upon accessing the platform. This main interface features a versatile Navigation Bar that functions as the primary gateway, allowing seamless access to various system modules and tools. From this dashboard, the User can directly interact with Profile Management to view account details or initiate updates to their personal information. The system also displays real-time alerts through the Notification feature, ensuring that the User is constantly updated on important activities or status changes. Together, these integrated components allow the HomePage to function as an efficient launchpad, organizing essential tools and information in a single, accessible view.

- Activity Diagram





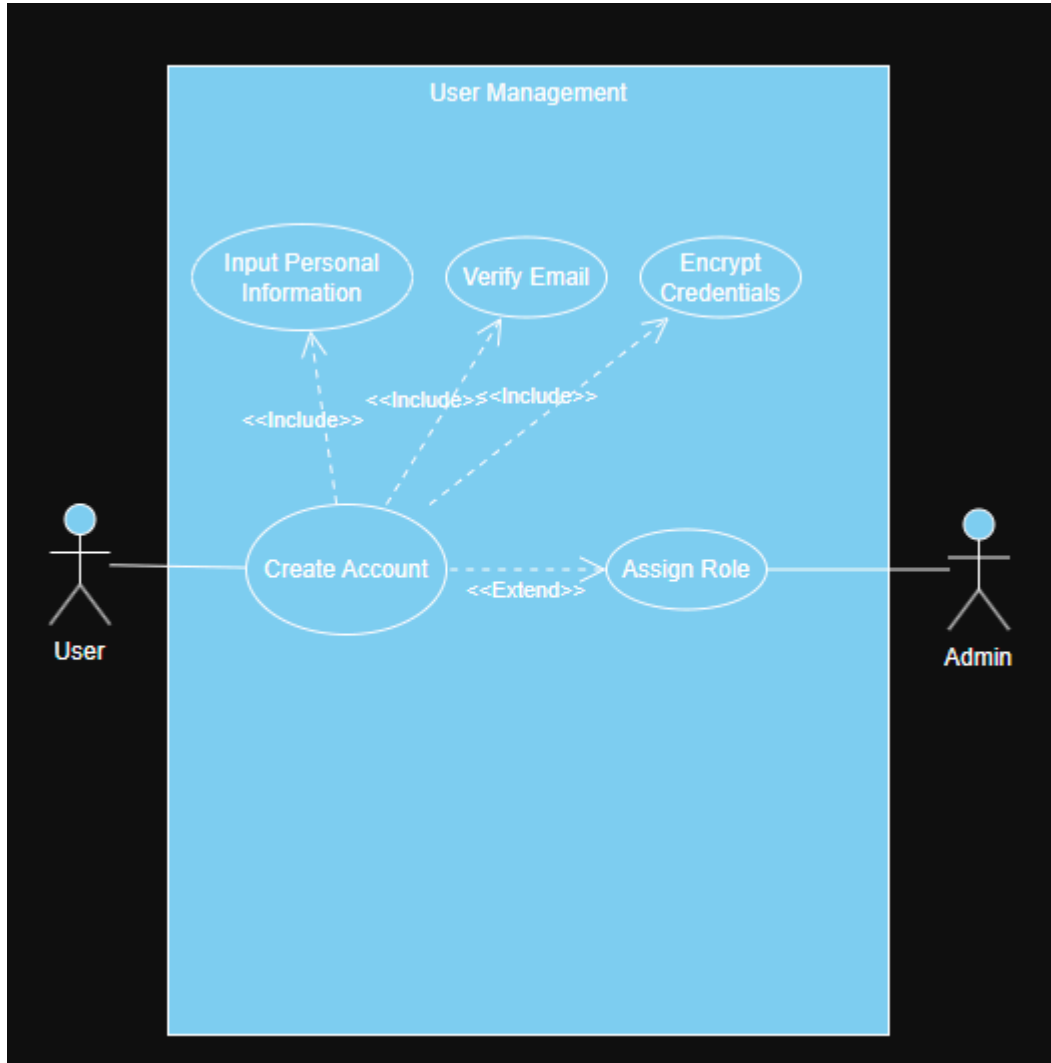
• Wireframe



## Module 2: Admin Panel

### 2.1 User Management

- Use Case Diagram

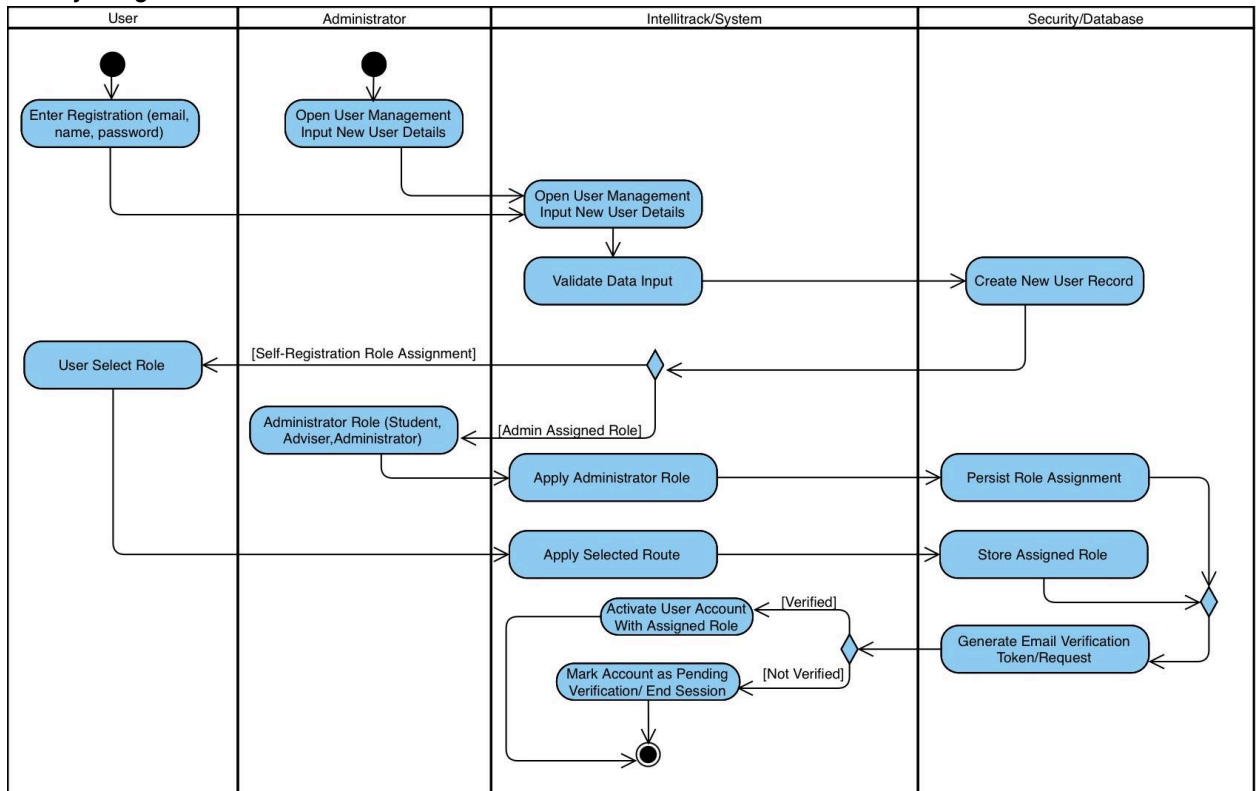


- Use Case Description

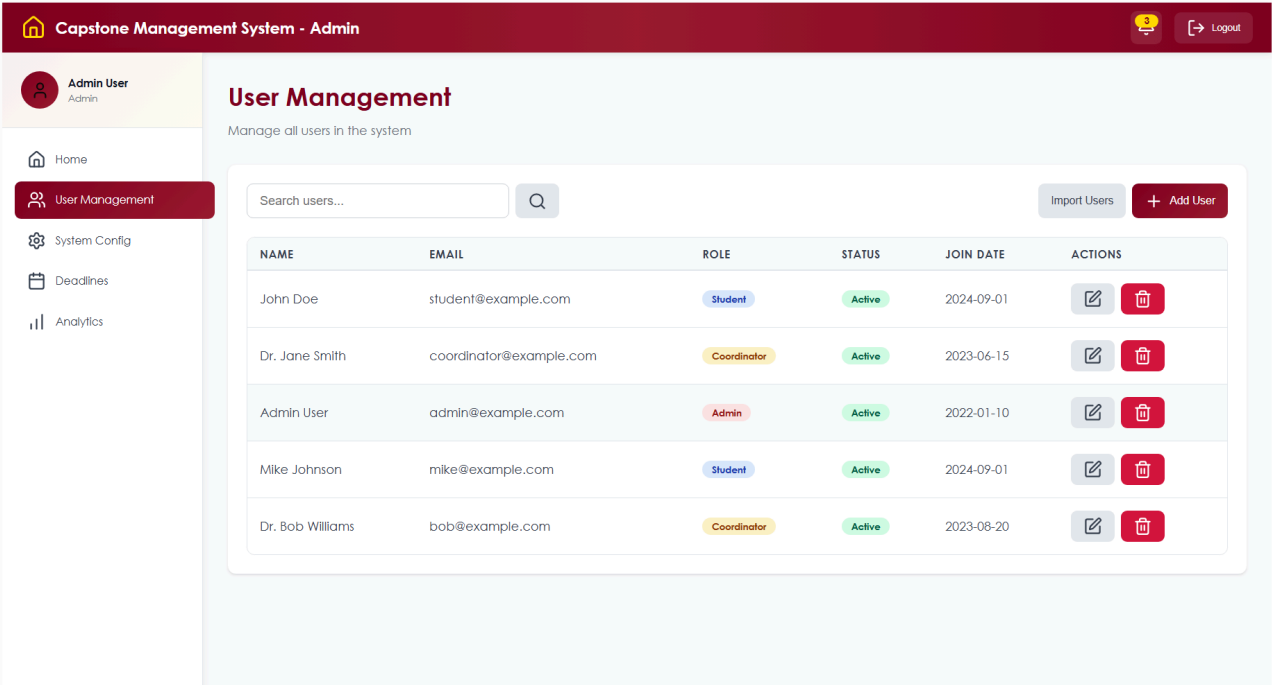
The Create Account use case governs the foundational process by which a new User gains initial entry to the IntelliTrack system by establishing a unique identity. Crucial to the system's security structure, this process directly facilitates Role-Based Access Control (RBAC) by explicitly assigning the User a specific designated role, such as Student, Adviser, or Administrator. While standard users may initiate self-registration by inputting their institutional email and personal details, the Administrator retains the authority to explicitly Assign Roles for faculty or staff members via the administrative interface. Upon successful account creation,

the system typically mandates a security verification step, such as email confirmation, to formally activate the User's privileges. Finally, this comprehensive process ensures immediate data protection through credential encryption, securely setting the stage for all subsequent user interactions within the platform.

- Activity Diagram



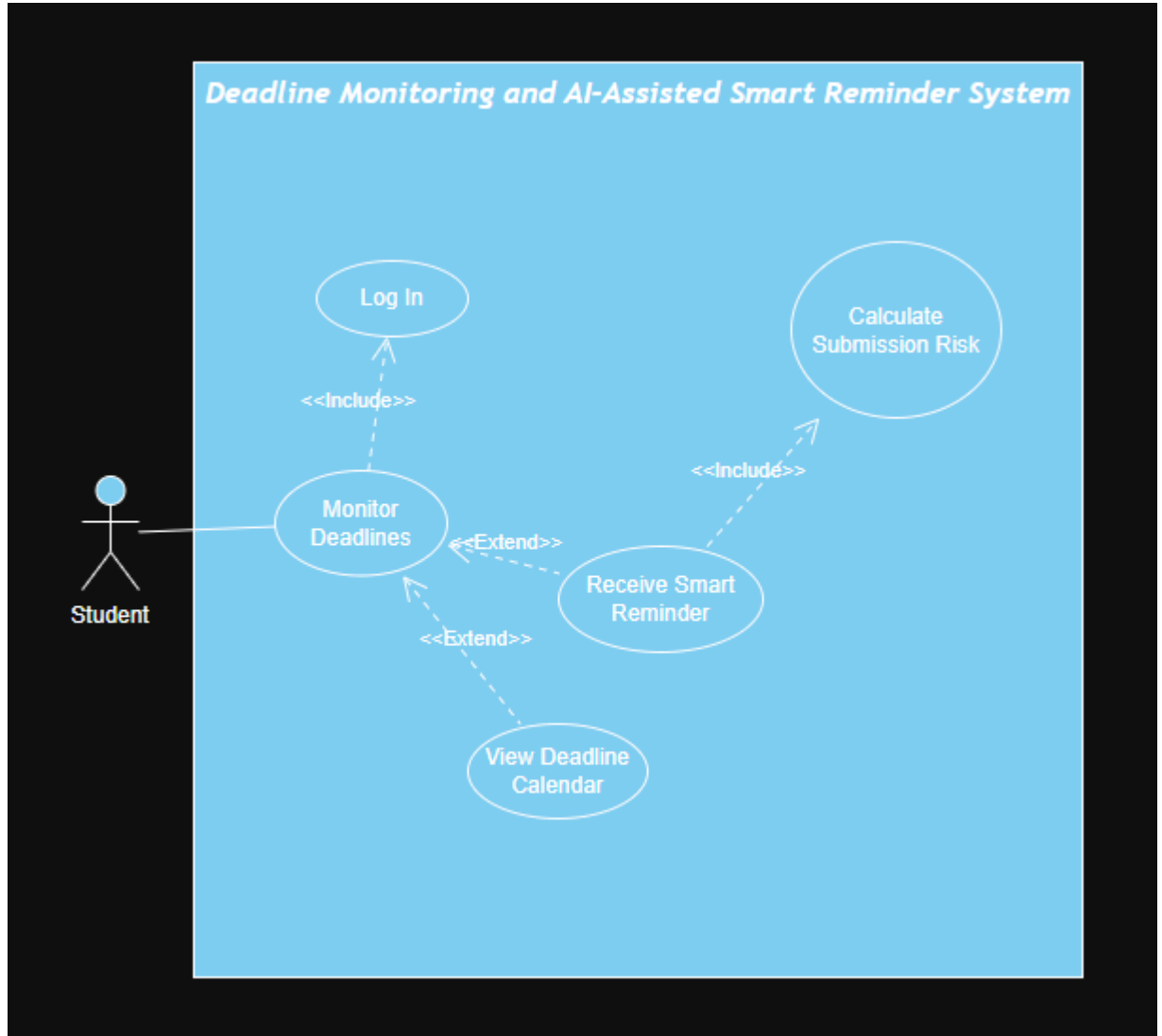
• Wireframe



## Module 3: AI-Powered Submission Tracker

### 3.1 Deadline Monitoring and AI-Assisted Smart Reminder System

- Use Case Diagram

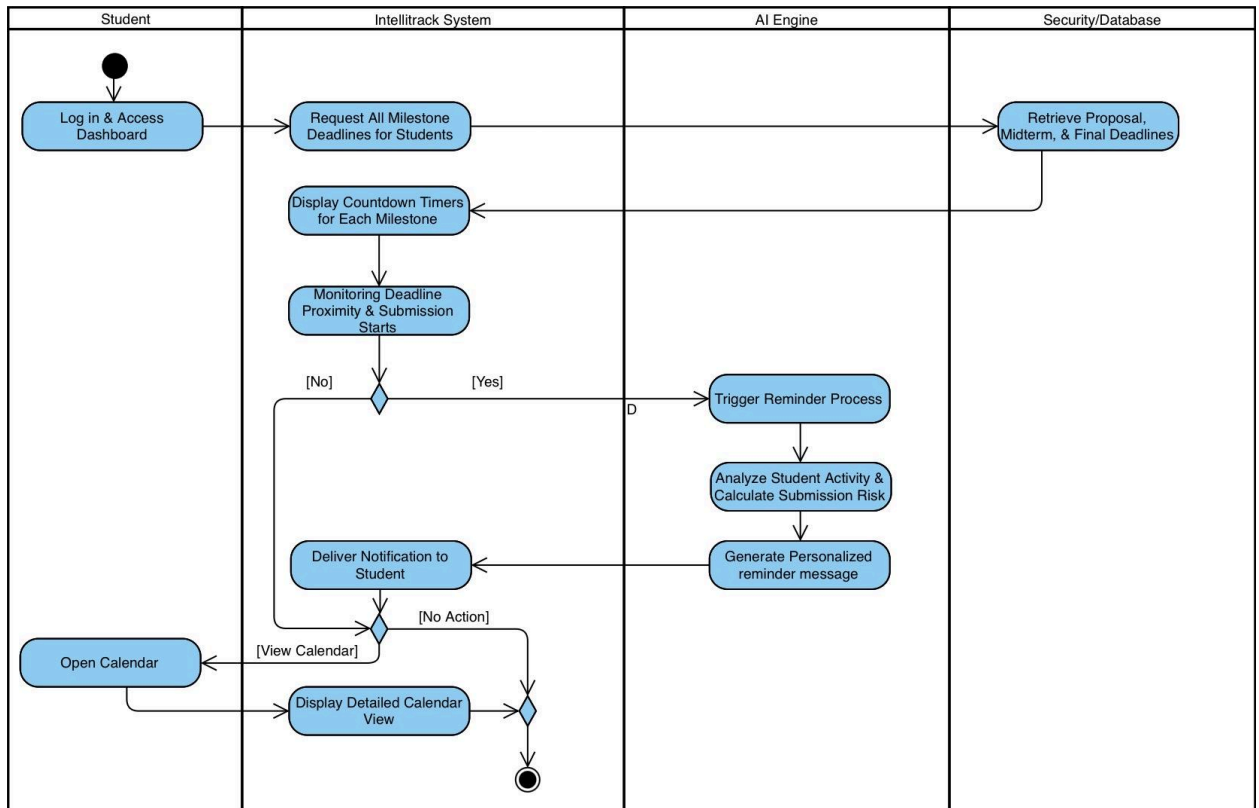


- Use Case Description

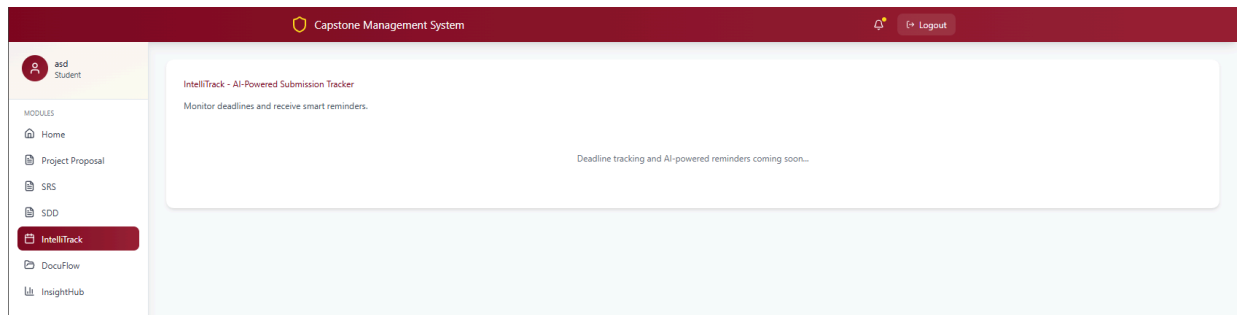
The Deadline Monitoring and AI-Assisted Smart Reminder System use case serves as the core of the AI-Powered Submission Tracker, which the Student accesses to manage their project timelines and receive necessary support. Upon logging in, the system retrieves and displays the student's deadlines, constantly monitoring these dates along with submission risk factors. This continuous monitoring conditionally triggers the Receive Smart Reminder function, which relies on the AI component to Calculate Submission Risk before delivering a notification.

The student also has the option to View Deadline Calendar for a detailed visual tracking of their submission requirements. This process effectively ensures the Student remains informed and supported in meeting all project deadlines through intelligent, automated assistance.

- Activity Diagram



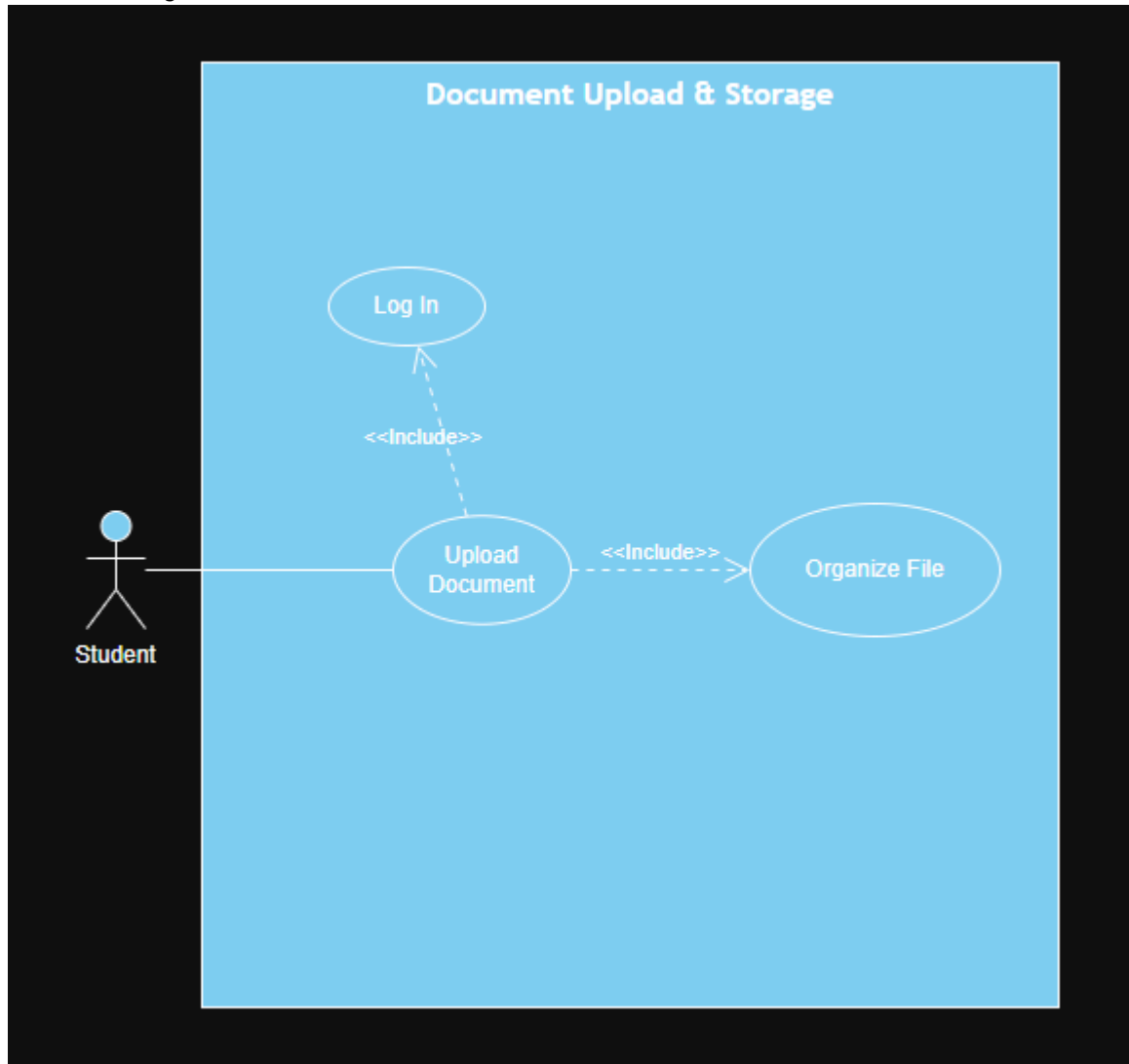
- Wireframe



## **Module 4: DocuFlow (Document Management, Version Control & Feedback System)**

### **4.1 Document Upload & Storage**

- *Use Case Diagram*

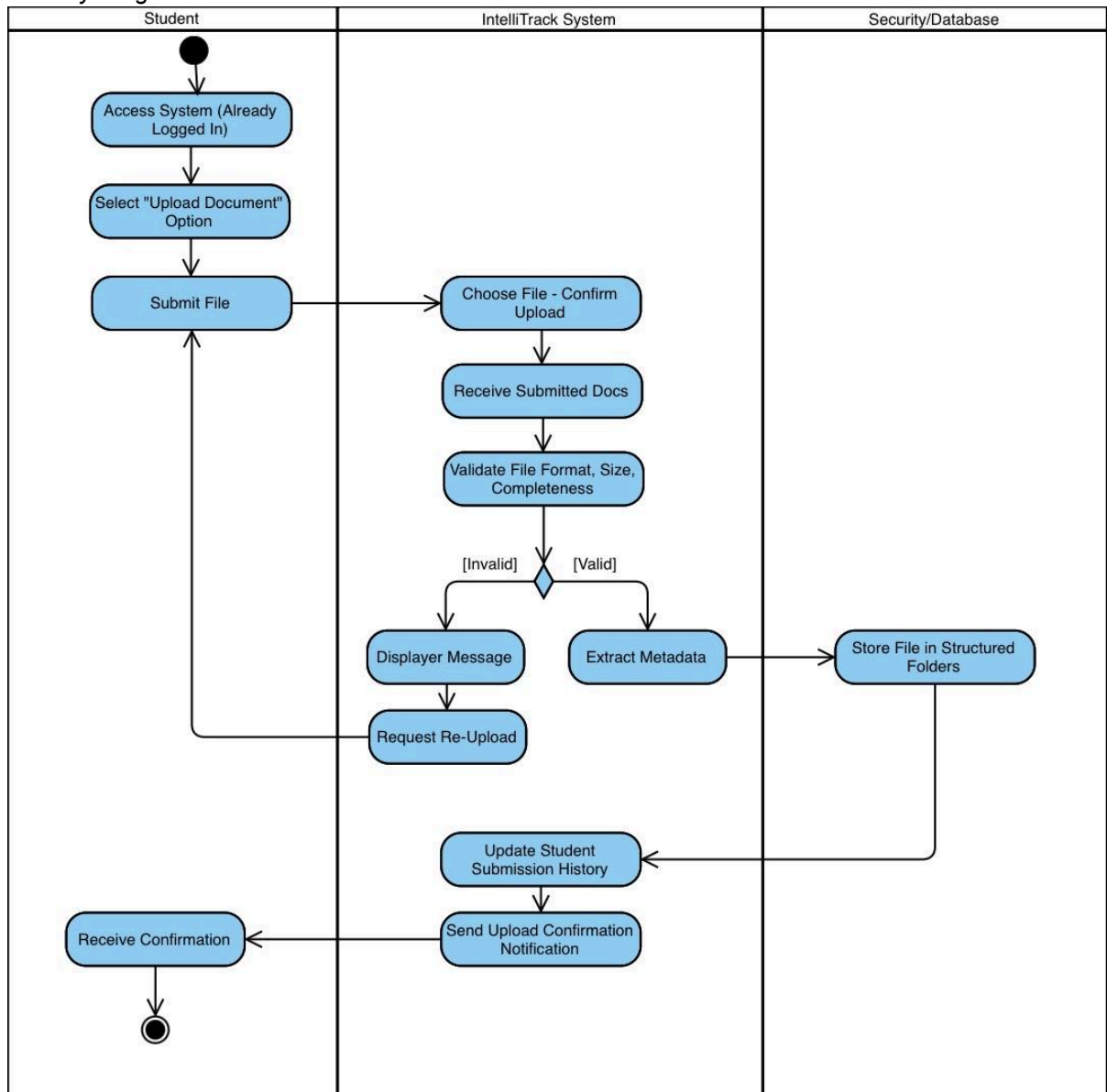


- *Use Case Description*

The Upload Document use case is the primary function for students, enabling the Student to submit project documents like the Project Proposal. To initiate this, the Student must first be successfully authenticated via the Log In process. Once the file is submitted, the system automatically handles the organization, versioning, and secure storage, which are inherent, built-in responsibilities of the document management system. This streamlined process allows the Student to focus solely on the submission task. This sequence ensures all project files are uploaded and

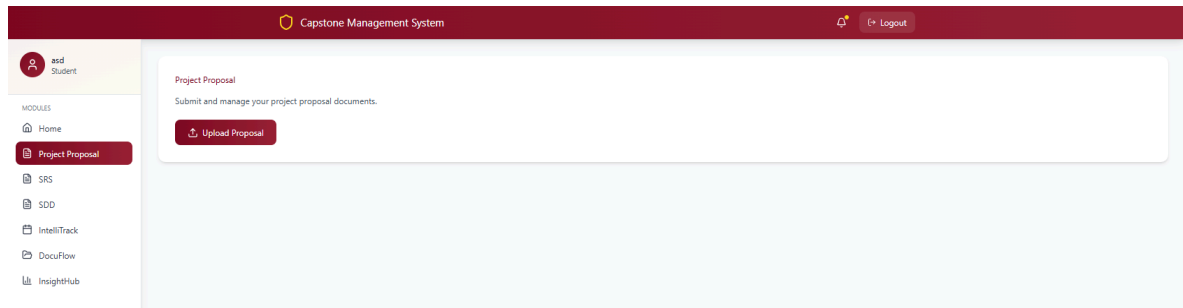
indexed correctly with minimal user complexity.

- Activity Diagram



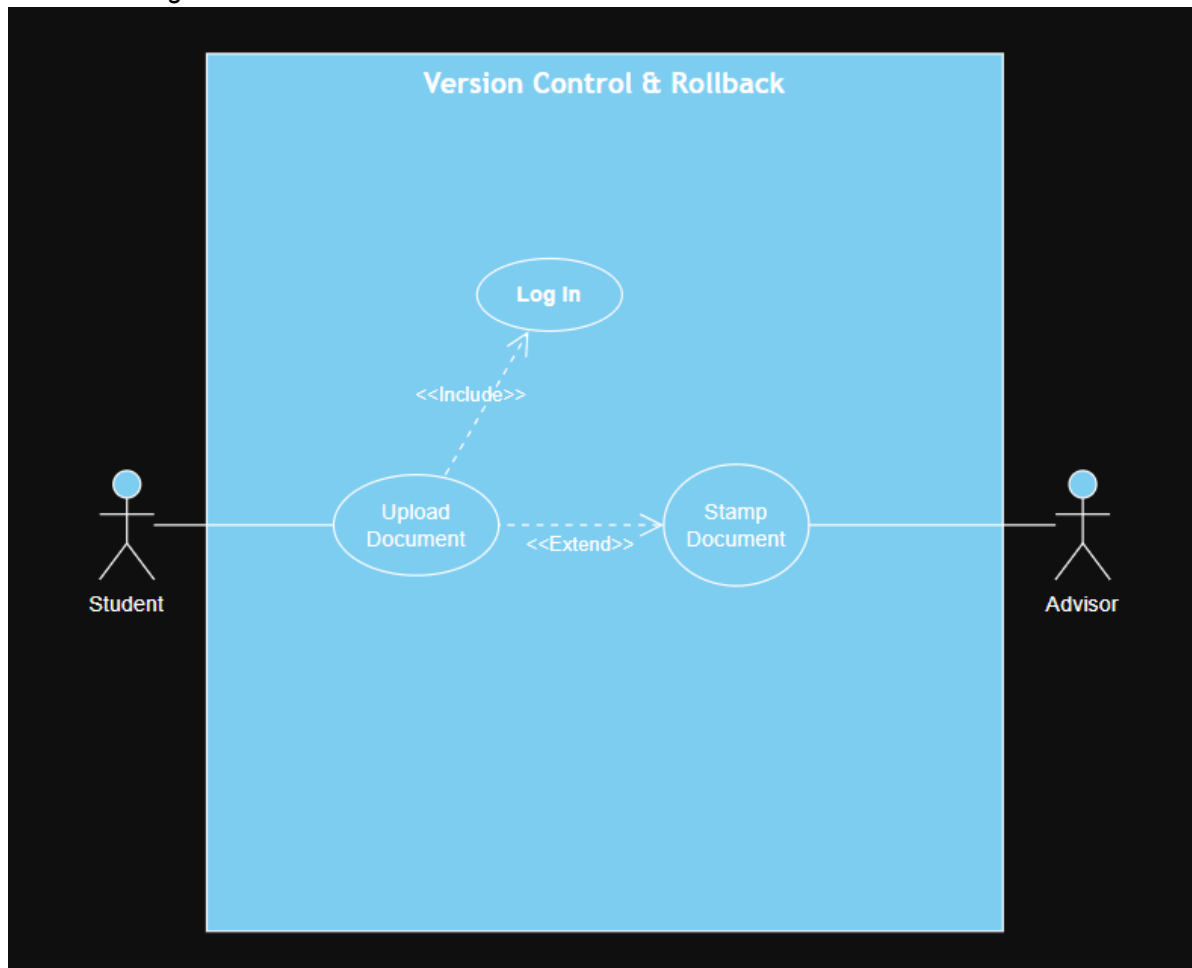


- **Wireframe**



## 4.2 Version Control & Rollback

- **Use Case Diagram**

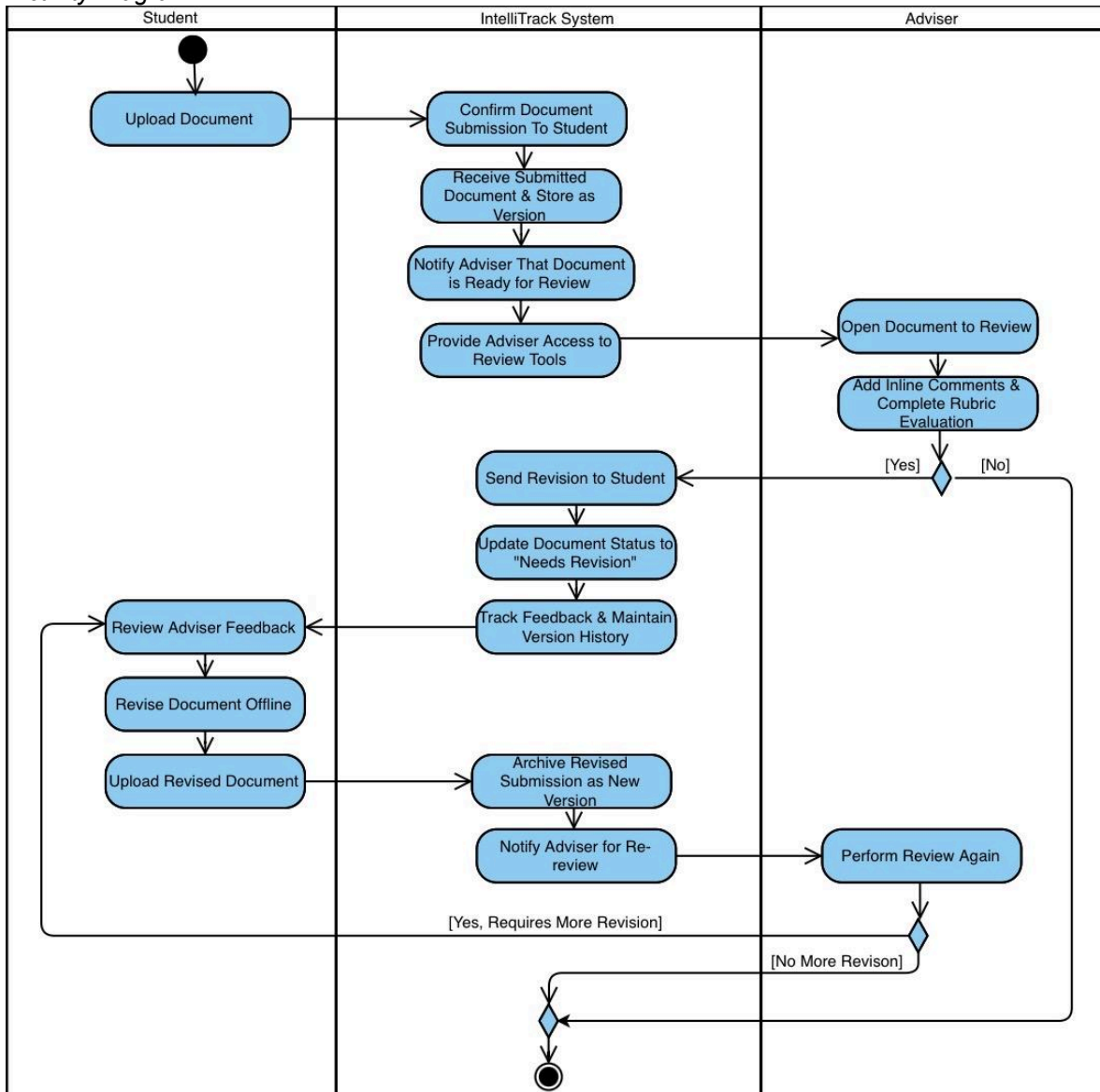


- **Use Case Description**

The Upload Document use case allows the Student to submit project documents like the Project Proposal after successfully completing the Log In process. Once the file is uploaded, the system internally handles the built-in organization and storage, making the document available for review. This submission then conditionally

triggers the Stamp Document function, which is exclusively reserved for the Adviser actor. The Adviser initiates this action to apply an official approval or rejection stamp to the submitted file, providing the necessary sign-off and controlling the document's flow within the system. This structured process enforces that all key project milestones are formally acknowledged and controlled by the supervising authority.

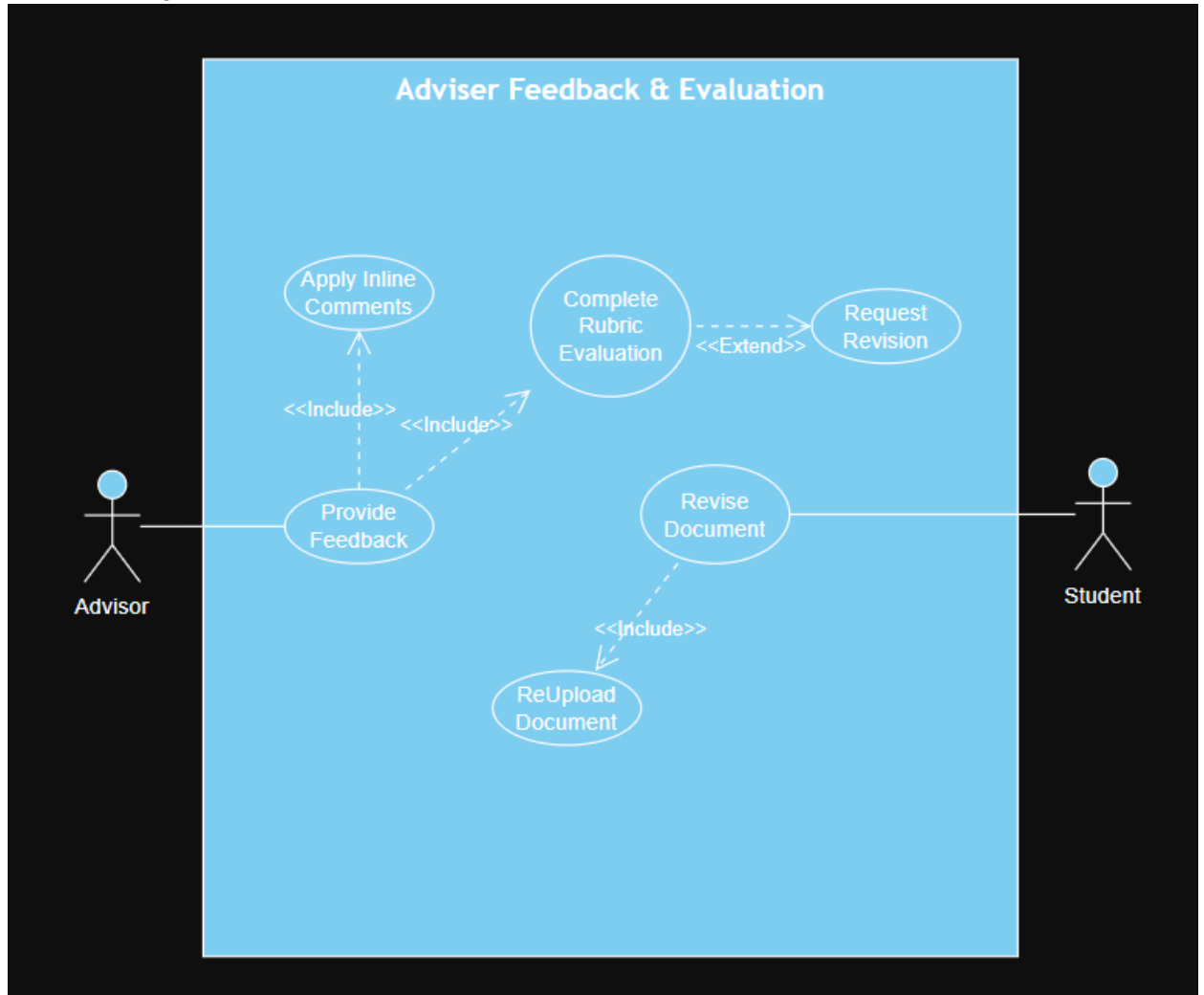
- Activity Diagram



- Wireframe

### 4.3 Adviser Feedback & Evaluation

- Use Case Diagram

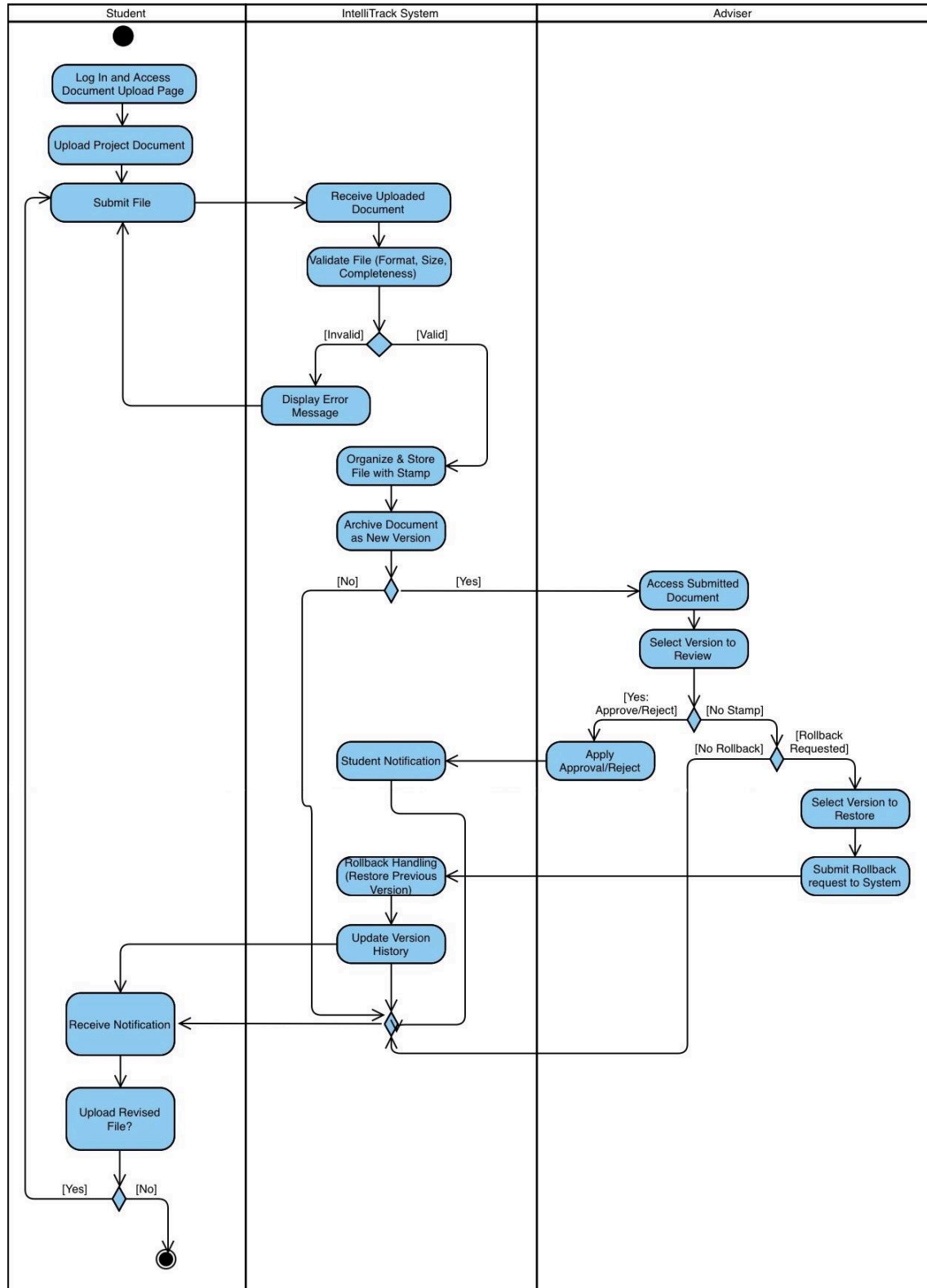


- Use Case Description

The Provide Feedback use case establishes the critical structured feedback loop within DocuFlow, initiated exclusively by the Adviser after a submission. To complete the review, the Adviser must use the mandatory functions of applying Inline Comments and Complete Rubric Evaluation. Following the evaluation, the Adviser conditionally triggers the Request Revision action, which formally shifts the responsibility to the Student. The Student then performs the Revise Document process offline, and upon completion, must use the existing Upload Document functionality to formally resubmit the revised file. This seamless integration ensures

the submission, review, and resubmission cycle maintains version control and clear audit trails throughout the project lifecycle.

- Activity Diagram



• Wireframe

Capstone Management System - Coordinator

3

Logout

Dr. Jane Smith  
Coordinator

Home

Calendar

Profile

Feedback Evaluation

Adviser Feedback & Evaluation

Review document and provide detailed evaluation

Save Draft

Submit Evaluation

Capstone Proposal - Version 3.0

Submitted by John Doe on Dec 16, 2025

Under Review

Document Viewer

Page 1

Add Comment

Document preview will appear here

Integration with PDF viewer or document renderer

Inline Comments (2)

Page 3

✓

✕

Need more citations here

Dr. Smith • 2025-12-16 10:30 AM

Page 5

✓

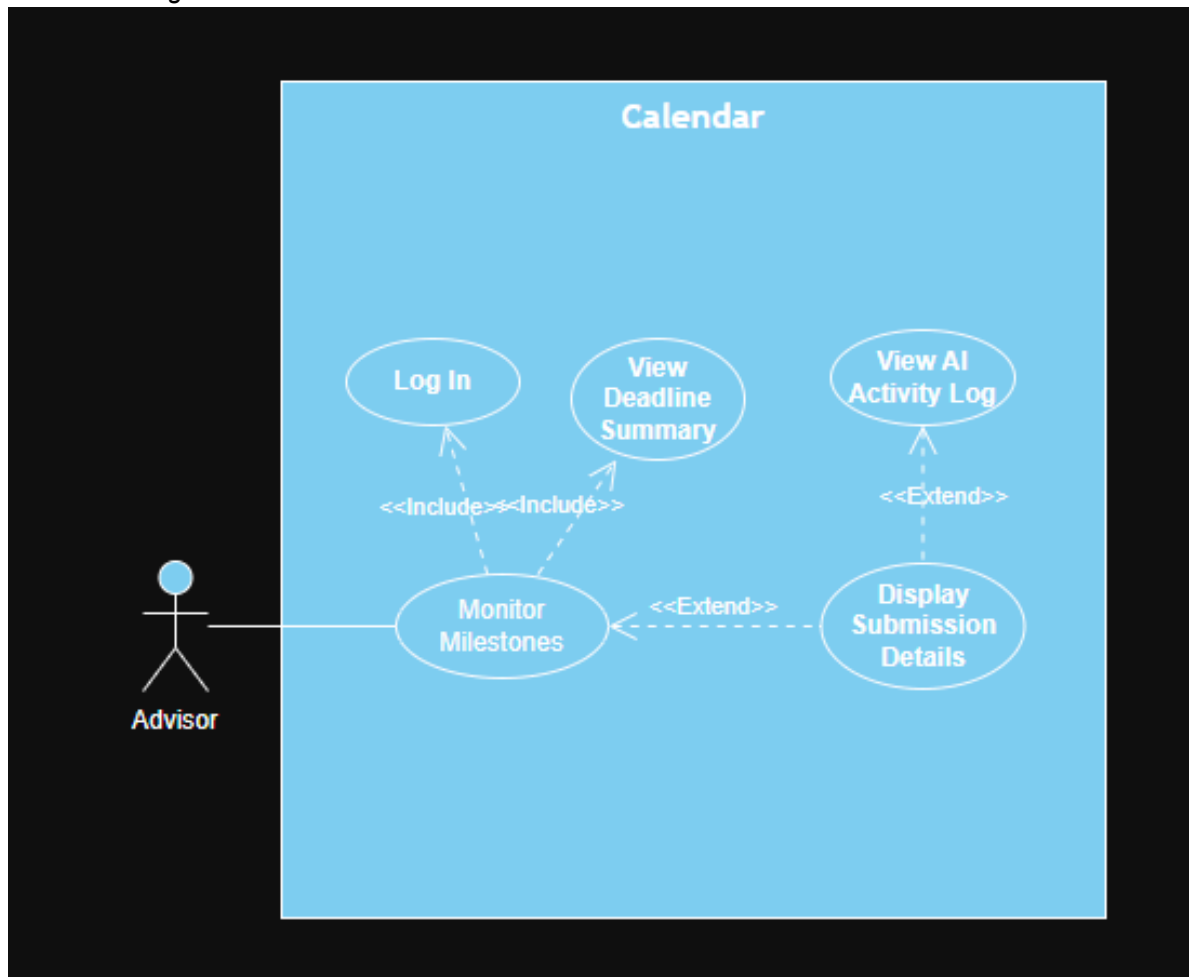
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Excellent analysis!

Dr. Smith • 2025-12-16 10:45 AM

#### 4.4 Milestone Monitoring & Interactive Calendar

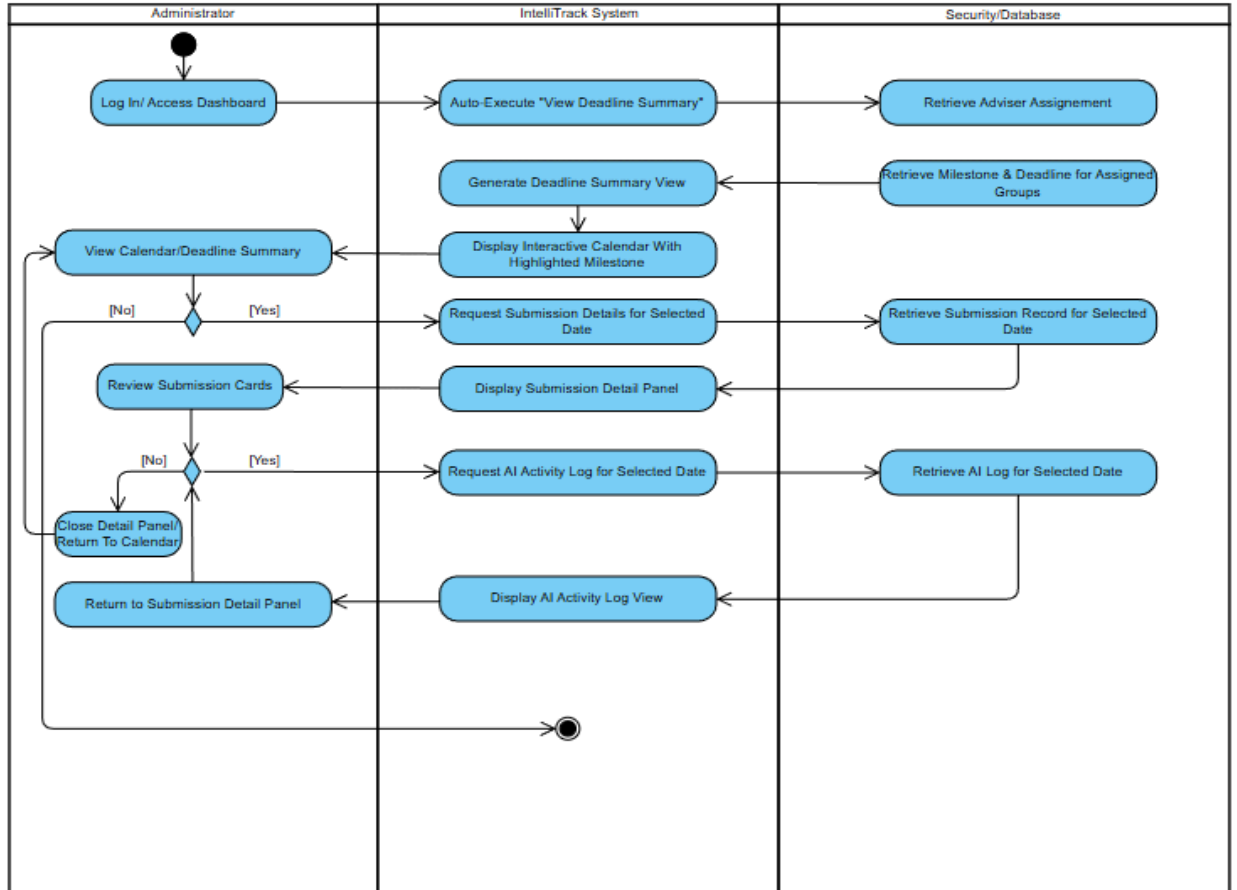
- Use Case Diagram



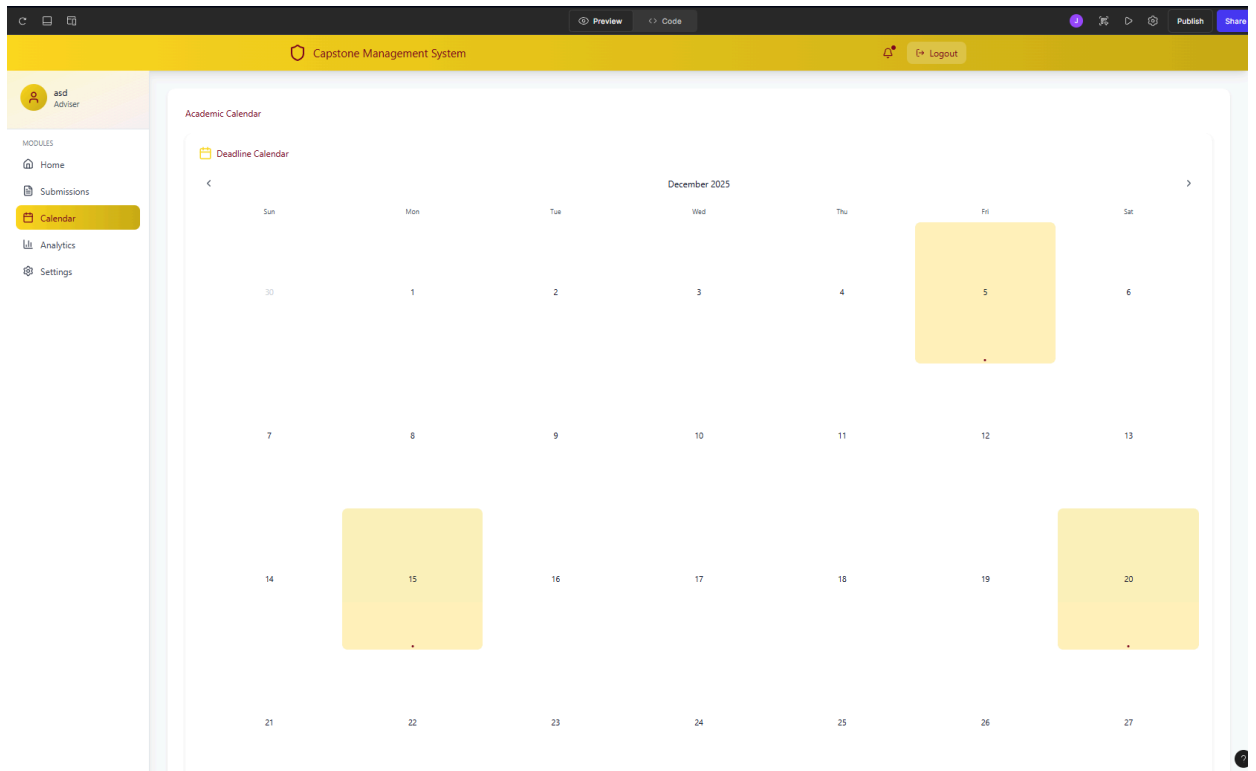
- Use Case Description

The Monitor Milestones use case provides the Adviser with a centralized view of project timelines via the Calendar. This process automatically executes View Deadline Summary, highlighting all milestones relevant to the Adviser's assigned groups, immediately upon Log In. The Adviser can then optionally trigger the Display Submission Details function by selecting a specific date, which opens an interactive panel showing submission cards with Group ID, Milestone Name, and Status. From this detailed view, the Adviser can further choose to View AI Activity Log, granting access to records of AI-suggested schedules and Smart Reminders sent to students on that date. This comprehensive visibility ensures the Adviser maintains constant oversight of project progress and AI-driven interventions.

- Activity Diagram



- Wireframe

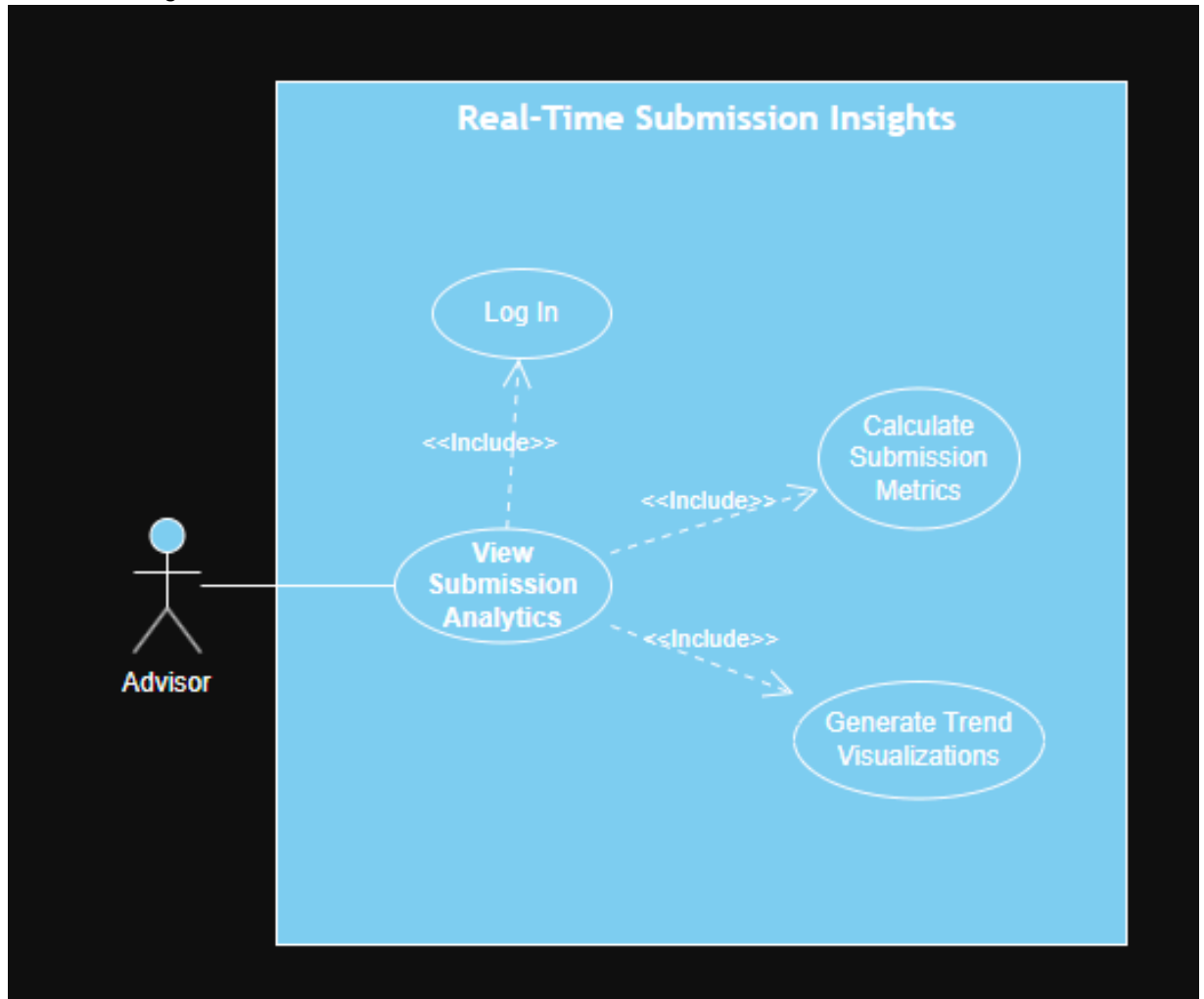




## Module 5: InsightHub (Analytics & Reporting Dashboard)

### 5.1 Real-Time Submission Insights

- Use Case Diagram

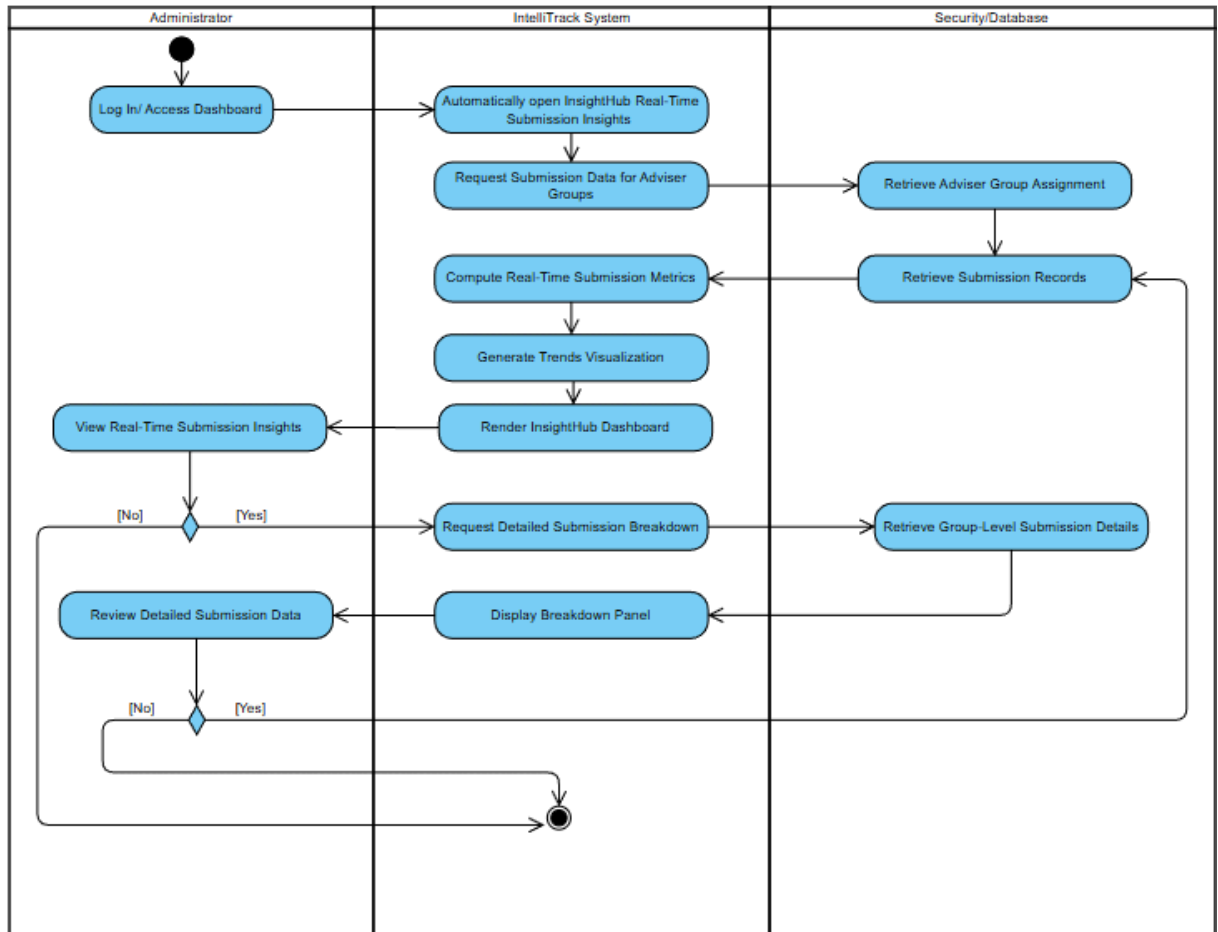


- Use Case Description

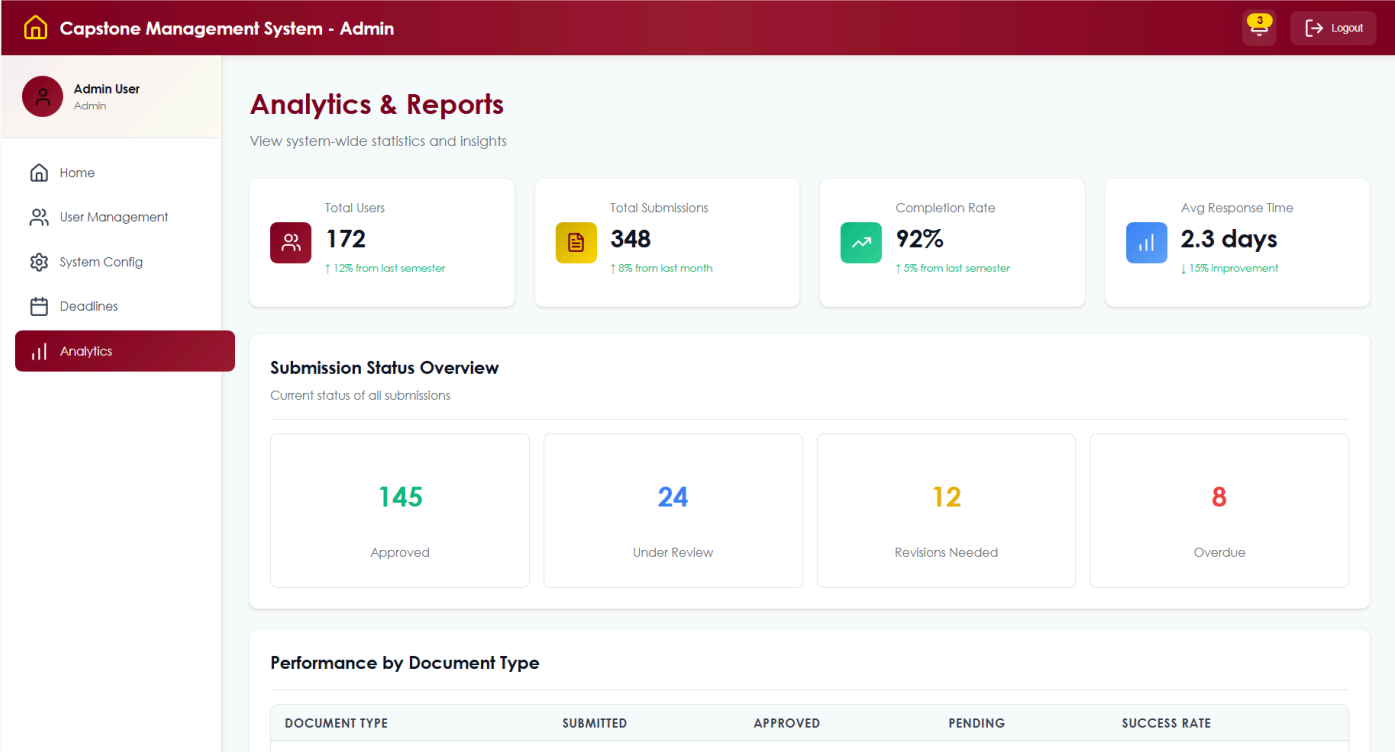
The View Submission Analytics use case provides the Adviser with immediate access to Real-Time Submission Insights within the InsightHub module. Upon successful Log In, this process automatically uses the Submission Metrics function, generating crucial data like the percentage of groups who have submitted on time, late, or are pending. Simultaneously, the system executes Generate Trend Visualizations, which translates project stage progress into easily digestible visual charts for the Adviser's review. This comprehensive and mandatory data processing ensures the Adviser has an accurate, high-level understanding of their

groups' performance and progress at all times. The entire module is designed to support proactive decision-making based on these instant metrics.

- Activity Diagram

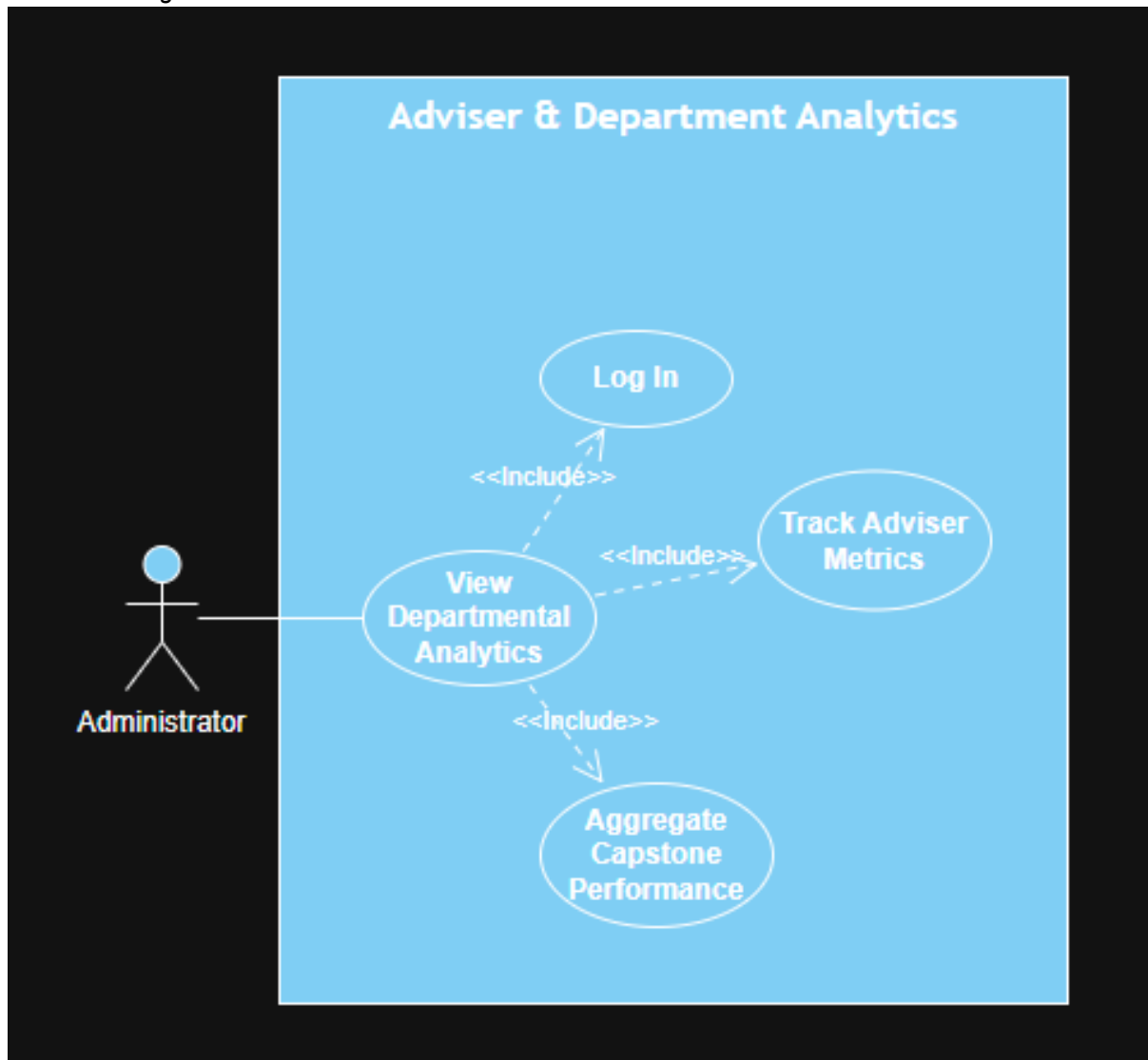


• Wireframe



## 5.2 Adviser & Department Analytics

- Use Case Diagram

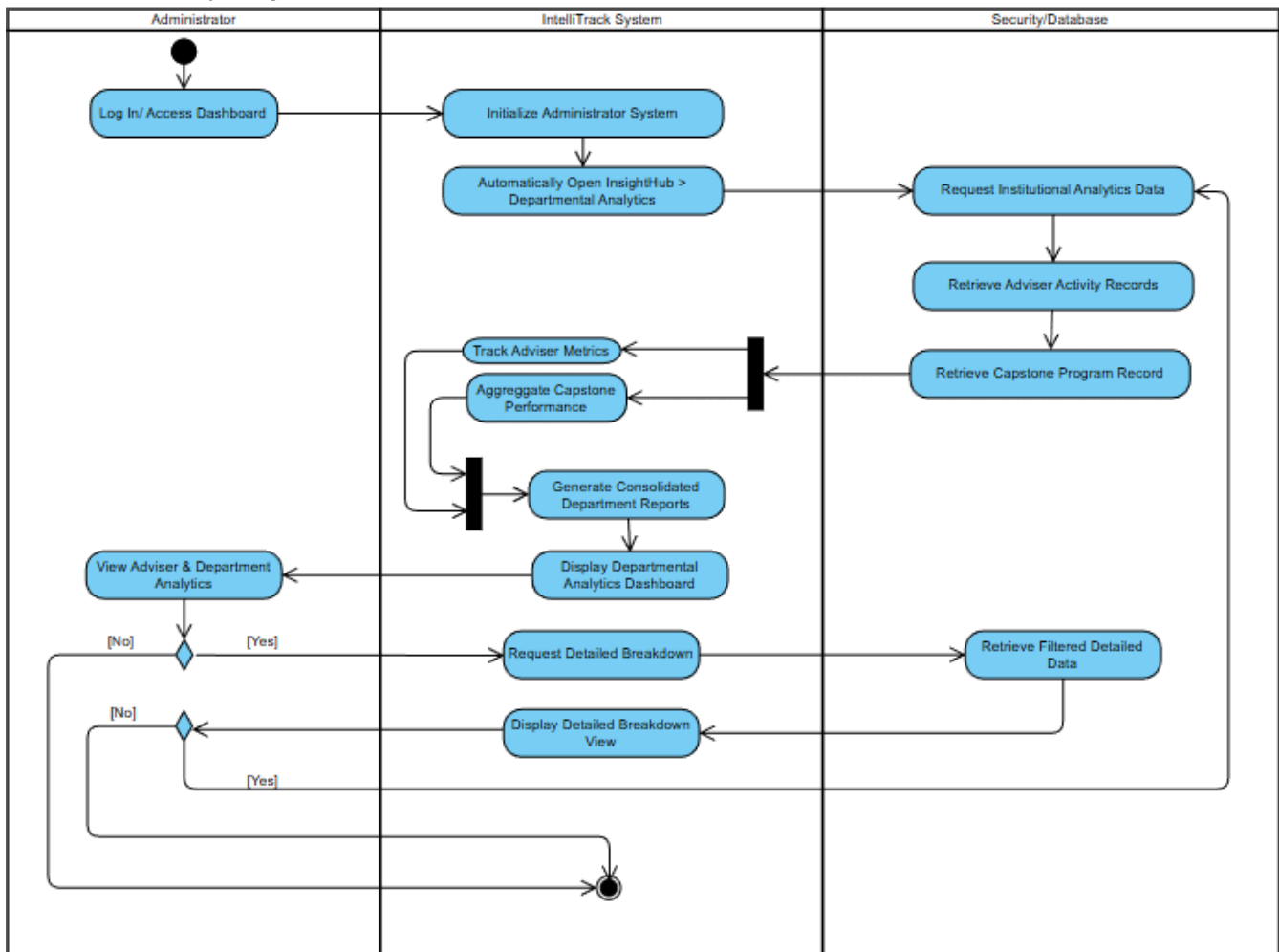


- Use Case Description

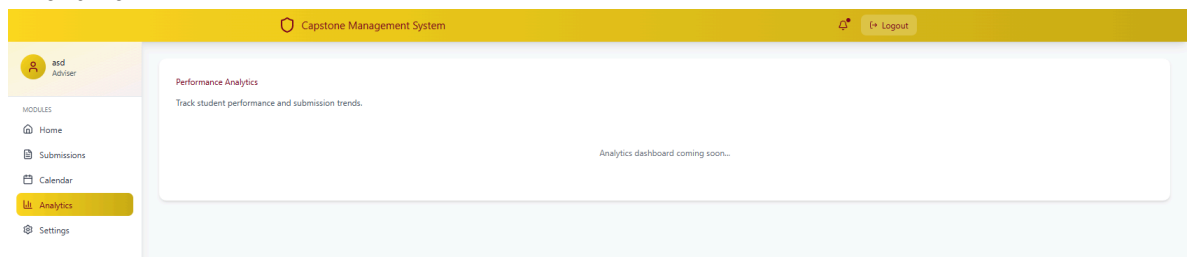
The View Departmental Analytics use case provides the Administrator with high-level reporting and institutional oversight tools within the InsightHub module. Upon successful Log In, the system is mandated to automatically execute Track Adviser Metrics, compiling real-time data on individual adviser performance, including their average response times and feedback quality ratings. Simultaneously, the system runs the Aggregate Capstone Performance function, which summarizes overall capstone data across all projects and groups for comprehensive institutional reporting. This constant, mandatory data processing

ensures that department heads and administrators have the necessary oversight to monitor staff efficiency and the overall health of the capstone program. The consolidated reports serve as a vital tool for making data-driven improvements to program management.

- Activity Diagram



- Wireframe



## ●.4 Non-functional requirements

### Performance

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- **Response Time:** All major user interactions (login, document upload, dashboard view) must be completed in **less than 2 seconds**.
- **Load Handling:** The system must handle **1,000 concurrent users** during peak usage periods (e.g., near deadline dates) without performance degradation.
- **Scalability:** The cloud-based architecture must support an increase in the number of users, projects, and future modules without requiring a system redesign or causing performance loss.
- **Real-Time Insights:** Analytics data displayed in the InsightHub must be refreshed in **near real-time** (data latency of less than 5 seconds).
- **Cross-Platform Performance:** System performance must be consistent and reliable across all supported major modern browsers (**Chrome, Edge, Safari, Firefox**) and desktop/mobile devices.

### Security

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- **Data Encryption:** All documents and sensitive user data must be protected using **end-to-end encryption** (both in transit and at rest).
- **Authentication: Multi-Factor Authentication (MFA)** must be enforced for all **Adviser** and **Administrator** accounts to enhance login security.
- **Access Control:** The system must strictly enforce **Role-Based Access Control (RBAC)** to ensure users can only access data and functions authorized for their specific role.
- **Compliance:** The system must adhere to institutional academic policies and comply with the **Data Privacy Act of 2012 (Philippines)** regarding user data and document handling.
- **Anomaly Detection:** The system must include mechanisms to detect and alert administrators to **suspicious login attempts** or unauthorized file activities.

## Reliability

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- **System Uptime:** The CDSS must maintain an **availability rate of 99.9%** (Platform Stability), excluding scheduled maintenance.
- **Data Integrity & Recovery:** The system must implement **automated backup and recovery plans** to ensure the integrity, consistency, and restorability of all submissions, feedback, and user data.
- **Auditability: Immutable audit trails** must be maintained for all critical actions (submissions, edits, role changes, and feedback) to ensure academic integrity.
- **Maintainability & UX:** The code base must be **modular, well-documented**, and follow established standards to ensure stability and ease of bug fixes. The interface must be **simple, intuitive, and clean** (Usability/UX) to minimize operational errors and increase user trust.